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Part Number: LM02460

ThermoSpot® Direct Contact Probe System (DCP - Series)

Operator's Manual

Revision A
May 2016

Preface

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Notice

Patents have been granted and/or patent applications are pending or are in process of preparation on all our developments.

The material in this manual is for informational purposes and is subject to change without notice.

inTEST Thermal Solutions assumes no responsibility for any errors which may appear in this manual.

To Our Customers

This Manual

The purpose of this manual is to help obtain the greatest return on your investment. inTEST Thermal Solutions suggests that operators, supervisors, and technicians responsible for operating and maintaining this equipment become familiar with the contents of this manual prior to using the equipment.

This manual instructs how to setup and operate the system. For service and maintenance, contact inTEST Thermal Solutions.

Support

inTEST Thermal Solutions is committed to assisting end users and technicians to maintain operational systems which are highly reliable. inTEST Thermal Solutions offers the following support services.

Customer Training

Formal technical training courses are available. The training courses cover the theory of operation and the maintenance procedures for the System. For further information, contact the inTEST Thermal Solutions Service Department.

Repair Service

inTEST Thermal Solutions maintains a fully equipped repair center at the factory plant for warranty and non-warranty repairs. For further information on module and circuit board repairs, our exchange program, and the availability of spare parts, contact the inTEST Thermal Solutions Service Department.

Before returning any module or circuit board for repair, contact the Service Department to obtain a return authorization (RA) number.

Spare Parts

Electrical and mechanical replaceable parts for the System can be obtained through your local inTEST Thermal Solutions representative, or directly from the inTEST Thermal Solutions Service Department. When ordering, be sure to specify the:

- Quantity
- Part number
- Description
- Reference designation (if any)
- Complete model number and serial number of your system

For your convenience, Spare Parts Kits are available for different levels of service activity.

Contact Support

Contact the Service Department by one of the following means:

inTEST Thermal Solutions Technical Support	
1.800.558.5080	Toll Free Telephone (service calls only)
service@inTESTthermal.com	e-mail Address
781.688.2300	inTEST Thermal Solutions Main Telephone

ATTENTION

1. Please note that the 1-800 toll free telephone number is dedicated to Service Department calls only. It is not possible to dial this number and to transfer to other departments within inTEST Thermal Solutions Corporation.

2. The main telephone number, 781.688.2300, should be used for non-service related calls.

Before You Call

You can help us support your machine in a timely fashion by having on hand specific information when calling in:

- Software Version
- System Model and Serial Number

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Safety

1

Warning Labels

The following labels and symbols may appear on the system:

WARNING

Carefully read the associated label. Refer to this manual for more information.
DANGER: Indicated an imminently hazardous situation which, if not avoided, will result in death or serious injury. DANGER is to be limited to the most extreme situations.
WARNING: Indicates a potentially hazardous situation which, if not avoided, may result in death or serious injury. It may also be used to alert against unsafe practices
CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in injury or damage to equipment. It may also be used to alert against unsafe practices

Electrical Hazard

High Voltage, Electrical Shock Hazard

Hot Surface

Hot Surface

Cold Surface

Cold Surface

WARNING: The locations of potentially dangerous voltages and other hazards such as hot surfaces or cold surfaces or compressed air/gases/vapors at high pressures are identified and labeled on the equipment. Be careful to observe these warnings when installing, operating, maintaining, or servicing the equipment. Observe all warnings given in this manual. Only use the equipment for the intended usages specified by the manufacturer.

WARNING: To avoid shock hazard, the equipment must be grounded with an adequate earth ground per local electrical codes.

WARNING: Setup, service and/or troubleshooting can involve possible contact with electrical power at high voltages, compressed air at high pressures, and refrigerant liquids at dangerous temperatures. These hazards can be injurious or dangerous to life. Only qualified service personnel familiar with the hazards present inside the equipment should perform any setup, disassembly, or corrective maintenance of the equipment.

WARNING: Where equipment must be fully de-energized (electrically “cold”) to allow safe entry into system, the following Lockout/Tagout procedure is required per OSHA 29 CFR 1910: a) For devices with a power cord which “unplugs” from a service receptacle, the end user must supply and tag a Plug lockout shell which completely encloses the plug and prevents accidental reconnection; b) for devices hardwired to the electrical supply, the end user must install and tag a power disconnect switch with a lock out position, or install and tag a circuit breaker with a lock out position, to prevent accidental reconnection.

WARNING: High Voltage is present inside the controller chassis. Remove power from the system before accessing the controller for service. Only a qualified electrical technician should remove the panels of the controller.

WARNING: If service of the refrigeration circuit is required, only a licensed/certified refrigeration technician should perform any charging or handling of the refrigerants in the system. Service by unqualified persons can cause warranties to be voided. Follow all Federal, State, and Local laws and regulations when handling refrigerants.

WARNING: Handle and dispose of any refrigerants, coolants, fluids and chemicals in accordance with the established policies and procedures for that material.

WARNING: Only use the coolants, refrigerants, and heat transfer fluids specified by the manufacturer, they are carefully engineered to be safe for operating personnel, to be friendly to the environment, to operate efficiently, and to be safe for the equipment. Do not substitute unauthorized coolants and refrigerants, nor mix (add) in unauthorized coolants or refrigerants: doing so can cause warranties to be voided. Wear protective safety eye glasses, gloves, and apron when filling coolants and refrigerants. Temptronic assumes no liability for damages caused by use of unauthorized coolants and refrigerants.

Cautions

CAUTION: Observe the precautions given on the equipment and within this manual to prevent damage to the equipment. Only use the equipment for the intended usages specified by the manufacturer.

CAUTION: Unauthorized personnel should not remove panels from the equipment, the panels are provided for protection.

CAUTION: Before connecting the equipment to its electrical source, check that the ~ (ac) voltage and frequency to be supplied to the system are correct for those listed on the system’s nameplate (located on the rear panel of the equipment).

CAUTION: Disconnect the system’s power cord from its service supply before checking or replacing any components.

CAUTION: The Flash Port on the I/O panel is a USB connection intended for factory service and diagnostics. Only a qualified Thermonics technician should use this port.

CAUTION: Do not block or restrict air flow at the condenser fans. Allow a minimum of 18” clearance at the fans.

CAUTION: The system should be returned to ambient temperatures before shutdown or service.

CAUTION: The (yellow) K-type thermocouple mounted to the thermal head is fragile. Do not bend the thermocouple wire beyond a 1/8” radius.

CAUTION: Do not tamper with the Allen Head (Hex Head) screws on the ThermoBridge. Adjusting this hardware may compromise the integrity of the thermal head and reduce performance. Only remove the Phillips Head screws to replace the ThermoBridge.

CAUTION: Do not twist or kink the thermal head and hose. The hose contains refrigerant lines that will be damaged if the head or hose is kinked. The head is designed to move freely and be positioned over the test area but do not force the head or hose if resistance is felt.

CAUTION: To protect your device under test (DUT) and test setup for moisture, condensation, and water, it is strongly suggested that you use purge air when testing at temperatures below ambient. Refer to the ThermoSpot data sheet for purge air supply requirements.

CAUTION: Purge air supply that is outside of the requirements listed on the ThermoSpot data sheet could adversely affect heating and cooling performance and fail to prevent moisture, condensation, and frost build-up.

CAUTION: The controller's setup parameters (F#s) give users access to controller settings that can adversely affect system performance if they are changed. Only a qualified technician or advanced user should change the setup parameters.

System Setup, Operation

2

System Description

ThermoSpot DCP-Series Systems are available in 4 models:

- DCP-101 - low capacity, US-domestic
- DCP-201 - low capacity, International
- DCP-102 - high capacity, US-domestic
- DCP-202 - high capacity, International

They are portable, temperature control systems that provide fast, accurate temperature for thermal testing of semiconductors, PCBs, and other electronic devices. The system consists of 4 main components: a refrigeration chassis, a touch-screen controller, a thermal hose, and a thermal head. Optional stands and mounts are available to hold and maneuver the thermal head over the test devices.

The thermal head features a conductive surface that efficiently heats or cools components once contact is made. The head is equipped with interchangeable ThermoBridge™ fixtures to match the geometry of the device under test (DUT).

Additionally, each system is equipped with a purge air system that circulates dry air at the test site. The dry air is used to reduce moisture and frost build-up that is common in low temperature testing applications.

In this Chapter

This chapter is divided into the following Sections:

Topic	See Page
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Section A: System Nameplate and Specifications

The nameplate is located on the system's rear panel. It lists specifications and critical information about the system.

SYSTEM INFORMATION	
Part#	
Model:	
Description	
Temp Range:	
Serial #:	
MFG Date	
ELECTRICAL	
Voltage:	
Freq:	
Phase:	
Current Draw Max:	
Min Circuit Ampacity	
Max Breaker	
Power Connection	

Section B: System Installation

Unpacking

The system is shipped on one container. When received the container should be examined for any signs of mishandling or damage during shipment.
NOTE: if there are any obvious signs of damage when the shipment is received, contact the carrier immediately and do not continue with unpacking or installation.

After the equipment is unpacked, retain and store the packing materials for future shipping of the system. Perform the following visual inspections of the system:

1. Verify that the controller touch screen is not scratched or broken.
2. Check the cabinet and refrigeration chassis for any scratches or damage.
3. Verify that the rear panel I/O ports are properly seated and the pins are not bent.

Placement

The system can be place for use on a bench, stand, or cart. The larger, high capacity systems are about 145lbs (66Kg), 24”D x 17.5”W x 17.5”H (610mmD x 445mmW x 445mm H). Check the system data sheet for exact weight and dimensions.

Allow 8” (203mm) of front and rear clearance to allow air flow through the system’s refrigeration chassis.

Interconnects

ThermoSpot systems require the following connections to be made before the system is operational:

- **Purge Air** - Purge air is used to prevent moisture and frost build-up. Purge is required for the thermal head (PROBE Purge) and optional for the test site (AUX PURGE).The inlets for both PROBE and AUX are located on the system’s rear panel. Purge Kit part numbers are:
  PROBE Purge (standard with system), includes the thermal head purge ring and purge collar, P/N SASA215460.
  AUX Purge (optional), includes tubing to connect facility air to rear panel purge inputs, P/N SA215280.

  Once dry air (or nitrogen) is supplied to the PROBE purge, no other connections are needed for the PROBE. Internal plumbing circulates the dry, purge air through the thermal head.
  The AUX probe requires a rear panel input connection and a front panel output connection. The AUX Purge outlet is located on the front of the chassis.

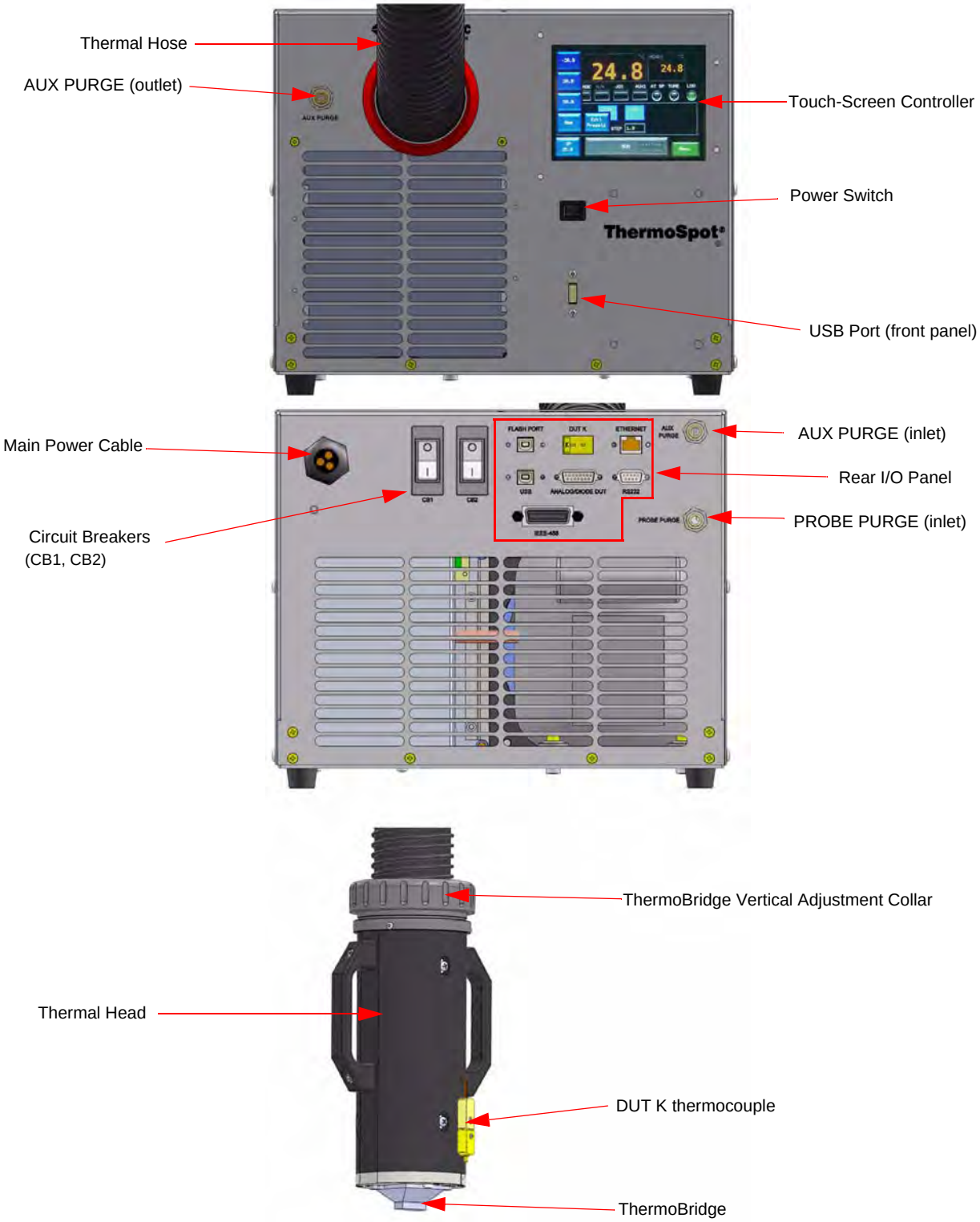
- **Power** - The system is equipped with a power cable configured for the voltage and frequency requirements specified at the time of purchase. 115 or 230 Volt, 50 or 60Hertz configurations are available.

The Importance of Purge Air

It is highly recommended that you connect purge air to this system. Purge air is critical for keeping the system and components being tested free of moisture an condensation caused by low temperature testing.

You can use your own tubing lines to connect purge air or order the Purge Air Kit from the factory. The kit is supplied with tubing, needle valves, and installation drawings.

Section C: Controls, Indicators, and Ports



Touch-Screen Controller

Lock / Remote Icons
lights when remote control or remote screen lock is active

DUT K
displays current temperature of DUT K

Soft Buttons
PURGE, turns on/off probe purge
JOG, turns on/off jog
AUX1 (optional) on/off switch

Set point (SP) button
Press to enter set point

Run / Stop button
Press to ramp and control at a set point or stop ramping and controlling

Warning / Error Icons
AT SP lights when the system is at

Head
displays current temperature of the thermal head

Status Lamps
REF lights when refigeration is on
AT SP lights when the system is at the temperature set point
TUNE lights when DUT autotune is on
LOG lights when data logging is on

JOG Buttons
+ and - to fine adjust temperature

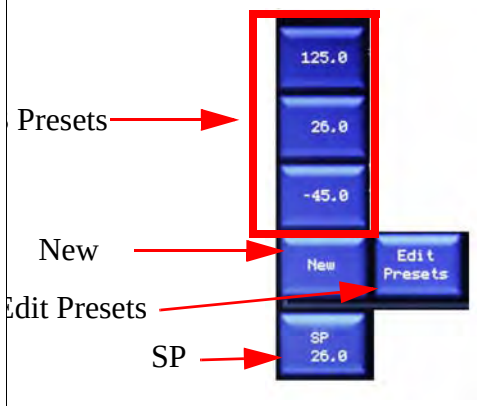
Menu button
Press to access controller menu.

Description of Controls, Indicators, and Ports

Control, Indicator, Port	Description
Thermal Hose	Connects the Thermal head to the chassis. Carries refrigerant to the thermal head.
AUX PURGE (outlet)	Outlet for auxiliary purge air. The auxiliary purge air is used to supply the test site with dry air to prevent moisture and frost build-up during low temperature operation.
Power Switch	Press to turn on/off the system
USB Port (front panel)	Type-A USB port for use with portable flash memory devices. This port can be used to connect external memory devices to store, save, or load programs and data logging.
Main Power Cable	The system's power cable is configured for the voltage and frequency requirements specified at the time of purchase. Connect the main power cable to the proper facility electrical supply. Refer to the system nameplate for voltage and frequency requirements.
Circuit Breakers (CB1, CB2)	Main circuit breakers for the system. CB1 is the main power breaker. CB2 is the compressor, heater, and valves breaker.
AUX PURGE (inlet)	Input for auxiliary purge air supply. Refer to the system data sheet for purge air supply requirements

Control, Indicator, Port	Description
PROBE PURGE (inlet)	Input for probe purge air supply. Refer to the system data sheet for purge air supply requirements
Rear I/O panel	Input / Output panel for additional connectivity to the system. **FLASH PORT**: a type-B USB Port for software and firmware updates. **DUT K PORT**: an optional connection for a Type K thermocouple - temperature sensor. Note: to use this port, the DUT K sensor at the thermal head must be disconnected. **ETHERNET**: a remote communications port for RJ45, 10/100 BaseT ethernet connection. Allows a remote computer to connect to the controller and control the system with IEEE-488 commands and queries **ANALOG / DIODE DUT**: a DUT sensor port for connecting an external temperature sensor. This port can accommodate a Diode, RTD, or analog signal (0 to 1V). **RS232**: (optional) a remote communications port for standard 25-pin RS232 Serial communication. Allows a remote computer to connect to the controller and control the system with RS232 commands and queries This Serial Port is configured as a DCE (Data Communications Equipment) and does not require a “null modem”. **IEEE-488**: (optional) a remote communications port for GPIB communications.
Thermal Head	The thermal head is designed to be positioned over and lowered onto the test devices. The thermal head uses conduction to heat and cool the device under test.
ThermoBridge Vertical Adjustment Collar	Used as a fine tune adjustment of the height of the ThermoBridge (up or down). The collar is typically used as a final adjustment once the ThermoBridge is in contact with the DUT. This final adjustment tightens the ThermoBridge to the DUT ensuring an ideal conductive surface between the bridge and the DUT.
DUT K thermocouple	A Type K (yellow) thermocouple sensor mounted to the thermal head. The DUT K sensor measures temperate at the bottom of the ThermoBridge, providing a precise, accurate temperature reading of the DUT.
ThermoBridge	A conductive, metal component that is installed at the bottom of the thermal head. The ThermoBridge makes direct, conductive contact with the device under test (DUT). It is designed to match the size and geometry of the DUT to ensure efficient conduction.
Touch-Screen Controller	
Lock / Remote Icons	Status icons that illuminate when the system is being controlled with remote communications and/or when the touch-screen is locked by a remote operator.

Control, Indicator, Port	Description
Warning / Error Icons	Status Icons that illuminate when there is a fault condition detected by the controller. Faults are either Errors (red X) or Warnings (yellow !).
DUT K:	Displays the temperature of the DUT K sensor. The DUT K sensor is a type K (yellow) thermocouple that reads the temperature at the tip of the ThermoBridge.
HEAD:	Displays the temperature of the RTD temperature sensor in the thermal head.
Soft Buttons	Rectangular shaped touch-screen buttons the turn system features on and off. The buttons illuminate green when in the on condition. **PURGE**: tuns on/off the PROBE PURGE. Note: Setup Parameter F38 (Purge Setpoint) must be configured for purge air flow. **JOG**: turns on/off the JOG+ and JOG- buttons on the touch screen. **AUX1**: (optional) can be configured to turn on/off an external device.
Status Lamps	Round indicators on the touch-screen. **REF**: illuminates when the refrigeration (compressor) is on. **AT SP**: illuminates when the system is at set point **TUNE**: (optional, for future release of tuning options) illuminates when DUT auto tuning is turned on. Note: Setup Parameter F7 (Tune Mode) is used to turn on DUT tuning. **LOG**: illuminates when datalogging is on.

Control, Indicator, Port	Description
Set point (SP) button	Press to access the set point flyout menu:  The set point flyout Menu contains: • **3 Preset Buttons** *(125.0, 26.0, -45.0 in the example above)*. Pressing a preset button loads that preset as the active set point. After the preset is loaded, use the RUN/STOP button to drive the system to active set point. • **New Button** Press to access the set point Screen. Using the New Preset button to access the set point screen allows you to change the active set point on-the-fly without changing or overwriting the (3) presets. • **Edit Preset Button** Press to access the Edit Preset Screen. The Edit Preset screen allows you to change the (3) presets. • **SP** Displays the active set point, i.e. the set point the system will drive to if you press RUN. This set point may be one of the (3) Presets or a set point entered on-the-fly. Pressing the Run button will run the system to the active set point.
JOG+, JOG- buttons	The JOG buttons provide an easy way to adjust the setpoint on the fly. To use the JOG buttons: 1. Press the rectangular JOG soft button to turn on the JOG feature. The JOG soft button will illuminate green. 2. Touch the JOG STEP window to adjust the amount the temperature will change with each jog adjustment. 3. Press the JOG+ or JOG- button to adjust the set point up or down as needed.

Control, Indicator, Port	Description
RUN (STOP) button	Press RUN and then Confirm Run to drive the system to the active set point. Press STOP and then Confirm Stop to stop the system from driving to the active set point. **WARNING** It is recommended that you return the system to ambient temperatures before stopping operations and/or shutting down the system.

Section D: System Startup and Shutdown

Introduction

Power and the optional, purge air connections must be made before starting up the system. A ThermoBridge should be installed at the bottom of the thermal head.

Turning ON the System

1. Verify that the main power cable is connected to a properly rated facility power outlet.(refer to the system's nameplate for voltage and frequency requirements).
2. At the system's rear panel, verify that CB1 and CB2 are in the ON (up) position.
3. At the system's front panel, toggle the power switch to the ON position.

   The system will turn on. It may take a few minutes for the refrigeration system to reach its full cooling capacity at initial startup.

Turning OFF the System

ATTENTION

To avoid leaving the thermal head at unsafe temperatures an/or shutting off the purge air at the site, the system should be returned to ambient temperatures and the thermal head removed from the test site before shutting down the system

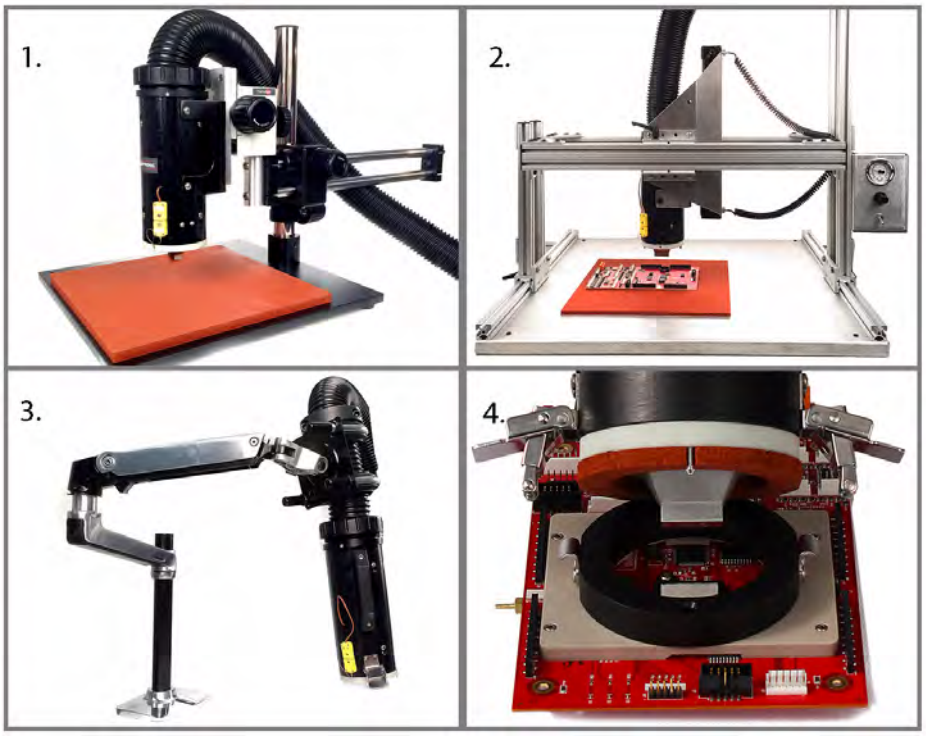
1. **Recommended.** Return the system to ambient temperatures. Do not leave the thermal head, or bridge at extreme temperatures.
2. **Recommended.** Disconnect the thermal head from the test site and remove the system from the device under test. Once the system is turned off, no purge air will flow and condensation could occur on any parts that are still cool. To protect the test site from moisture, move the thermal head away from the test site.
3. At the system's front panel, toggle the power switch to the OFF position.
4. Be sure your DUT is powered down.

Section E: Interfacing the Thermal Head with the DUT

Introduction

The thermal head is equipped with handles and can be manually moved and placed onto the device under test (DUT). Fixturing and stand options are available:

Stands and Fixtures



1. Boom Stand: provides X, Y, Z positioning of thermal head.
(p/n SA214350)

2. Pneumatic Stand: provides X, Y, Z, positioning and measured downward force to DUT.
(p/n SA214920)

3. Bench Stand: free-motion stand and boom to manually position the thermal head over DUT.
(p/n SA214360)

4. DUT Interfacing

- Test Socket Alignment - latches and guide pins provide easy and secure connections to the test socket.
- Soldered Component Alignment - customized interfaces to accommodate PCB layout and chip geometry.

Boom Stand Setup

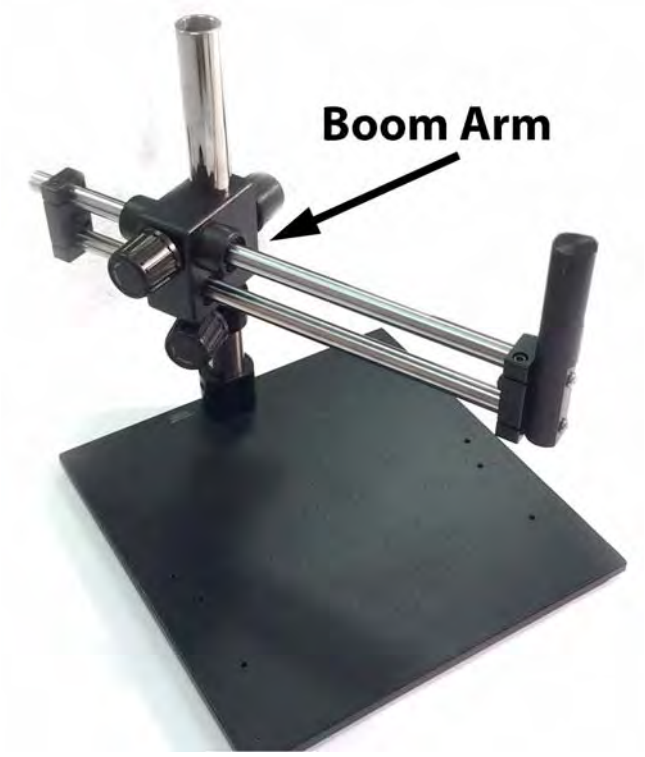
The (optional) Boom Stand provides X, Y, Z positioning of thermal head. It is shipped in a separate container. Unpack the Boom Stand and setup as follows:

Boom Stand Components

Setup Instructions

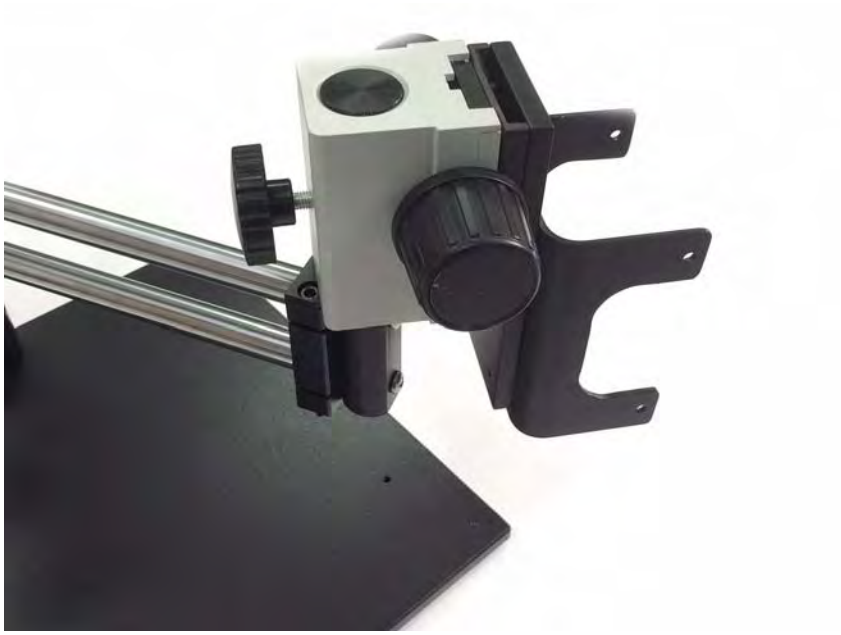
1. Slide the Collar Lock over the Base Plate Post.
   Note: use the knob on the Collar Lock to lock in place as needed.

2. Slide the Boom Arm over the Base Plate Post:

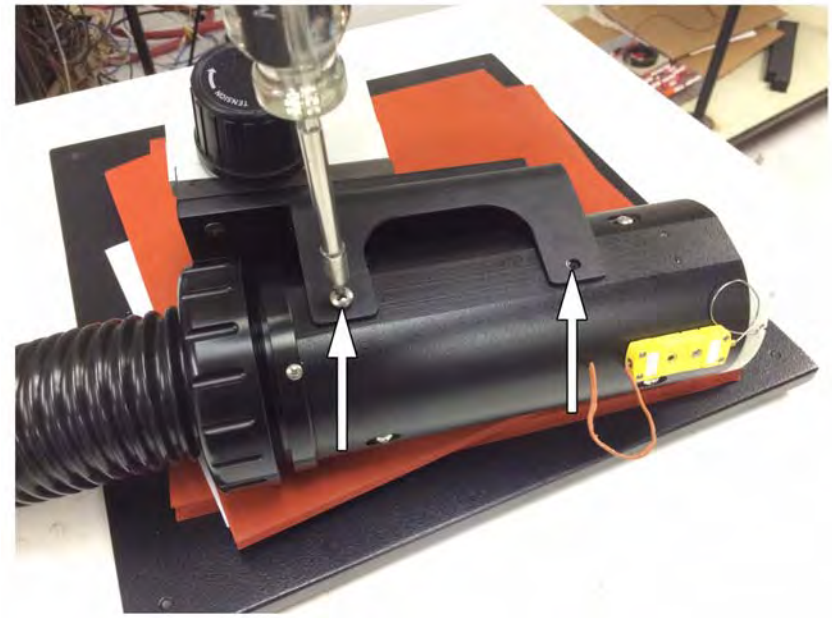


3. Insert the Cap into the post:

4. Install the Head Adapter onto the Boom Arm as shown:



5. Secure the Thermal Head to the Head Adapter using (4) 8/32 X 3/8”, stainless steel, screws.



6. The Boom Stand is no assembled and the Thermal Head is mounted.

Section F: Changing The ThermoBridge™

Introduction

The system is typically equipped with a ThermoBridge that meets the size and geometry of your DUT as specified at the time or purchase. If a ThermoBridge of different size or geometry is needed, it can be ordered from the factory and installed.

Follow this procedure to remove and replace a ThermoBridge.

ThermoBridge Removal

Before performing this procedure, return the system to a safe (ambient) temperature and disconnect power and air supplies from the system.

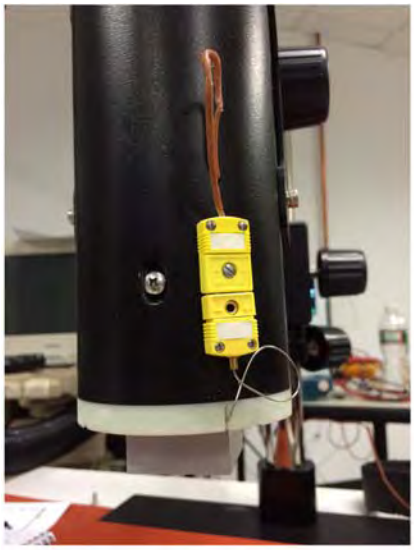
The ThermoBridge and (yellow) DUTK sensor are removed as one assembly.

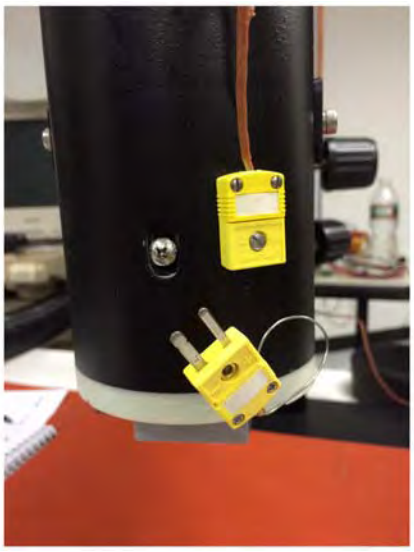
ATTENTION

Thermocouple wires are fragile, be careful handling and removing the thermocouple.
Do not bend beyond a 1/8” radius.

REPLACEMENT THERMOCOUPLE: Part Number RT00230

1. At the thermal head, use a flat head screwdriver to remove the hardware securing the (yellow) DUTK thermocouple. Pull the top (female) and bottom (male) connectors apart shown:

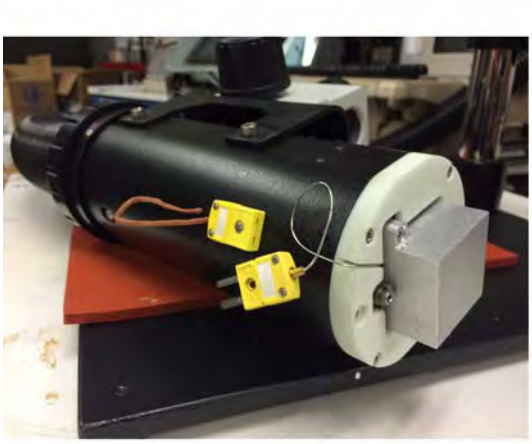


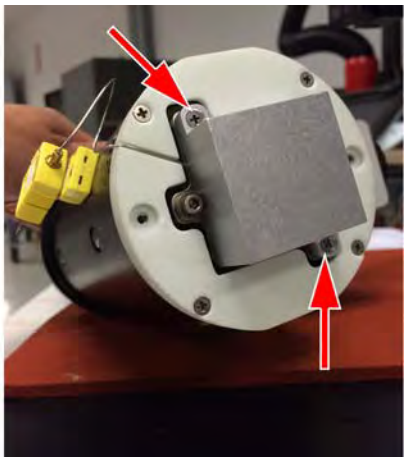


2. Carefully lay the thermal head down to gain access to the ThermoBridge and the hardware securing it.
3. Loosen the (2) Phillips Head Screws securing the ThermoBridge.
   Do not remove the ThermoBridge without holding the (yellow) DUTK thermocouple.

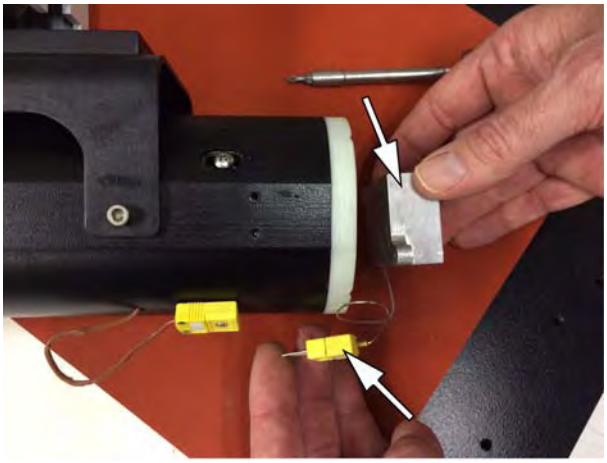
ATTENTION

DO NOT tamper with the Allen Head (Hex Head) screws on the ThermoBridge. Adjusting this this hardware may compromise the integrity of the thermal head and reduce performance. Only remove the Phillips Head screws to replace the ThermoBridge.





4. Carefully remove the ThermoBridge and thermocouple as on assemble.

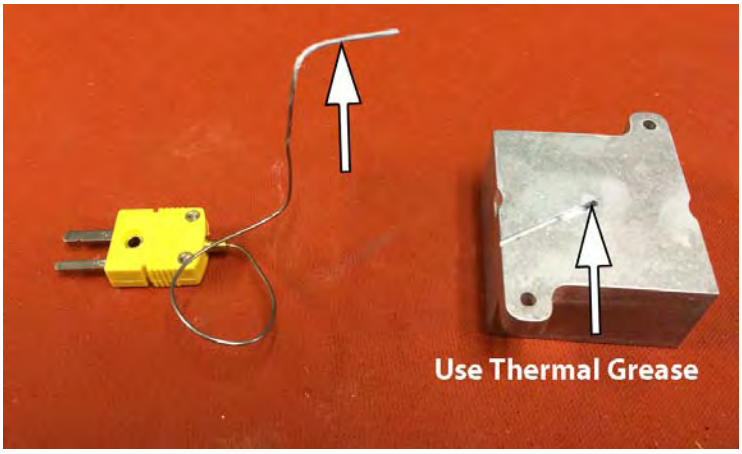


5. The ThermoBridge and Thermocouple are now removed.

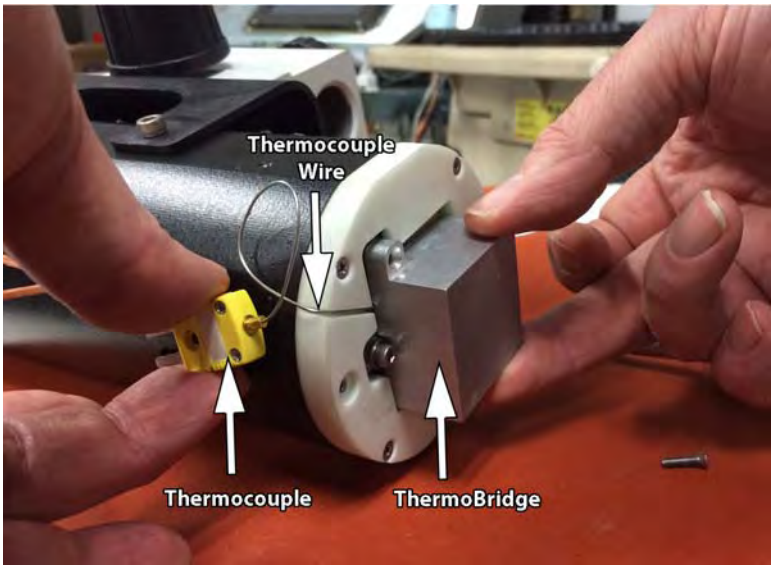
Installing a New ThermoBridge

Refer to the ThermoBridge removal procedure above as needed. Thermocouples can be ordered separately from the the ThermoBridge. Thermocouple part number: RT00230.

1. Use thermal grease (example: Boron Nitride Heat Sink Grease) to coat the thermocouple wire and fill the hole in theThermoBridge.



2. Insert the thermocouple wire into the ThermoBridge. Seat the wire as far into the ThermoBridge as possible.
   NOTE: If the thermocouple is not seated all the way into the ThermoBridge, it will adversely affect temperature readings.
3. Install the Bridge and thermocouple into the thermal head.
   - route the thermocouple through the channel in the thermal head
   - secure the thermocouple to the thermal head using the flat head screw.
   - secure the bridge to the thermal head with (2) Phillips Head Screws.



4. The new ThermoBridge and thermocouple are now installed. Resume normal operations.

Section G: DUT Sensor Setup

Connecting an external DUTK, DUTR, DUTD, or DUTA Sensor

Introduction

The system can control the temperature based on input from an external temperature sensor. The rear I/O panel provides inputs for the following sensor types:

- K Type thermocouple (DUTK)
- RTD (DUTR)
- Diode (DUTD)
- Analog Signal (DUTA)

DUT Setup Parameters (F#s)

To use an external sensor, the following setup parameters must be configured in the controller. To locate the setup parameters on the touch-screen:

MENU > Go SETUP > Setup Parameters

- F1: Number of probes installed, set to 2
- F35: Second Probe Type, choose DUTK, DUTR, DUTD, or DUTA
- F40: Temperature Control Mode: set to DUT

NOTE: If DUT Mode is not required, setting F40 to Normal will allow the system to control based on the RTD in the Thermal Head. This temperature is displayed on the touch screen as HEAD: .

DUTK (located on Thermal Head)

The yellow, DUTK sensor located on the thermal head is a K-type thermocouple used as the default sensor. This DUTK measures temperature at the ThermoBridge, providing the most accurate temperature reading at (or near) the DUT.

DUTK (located on rear panel)

To use the rear panel DUTK, the HEAD DUTK must be disconnected.

DUTR

A 3-wire, 100 Ohm, 385 curve RTD can be used as the DUT Sensor.

Refer to DUT pin outs diagram below.

DUTD

A Diode can be used as the DUT Sensor. The For accurate temperature readings, the diode ideality must be close to 1.0, anything over 1.25 may not be accurate.

Setup parameters F52 and F53 must be configured to match the diode being used.

Refer to DUT pin outs diagram below.

DUTA

An analog signal input. Requires a low impedance, 0 to 1 Volt signal. Setup Parameters F45 (Analog In Low Temp) and F46 (Analog in High Temp) must be configured to use DUTA.

For example:

- Set F45 (Analog In Low Temp) to 0 (zero degrees). An input signal of 0 Volts will be read as 0 degrees.
- Set F46 (Analog In Low Temp) to 100. An input signal of 1 Volt will be read as 100 degrees.

F45 and F46 may be set to any temperatures within the system's temperature range.

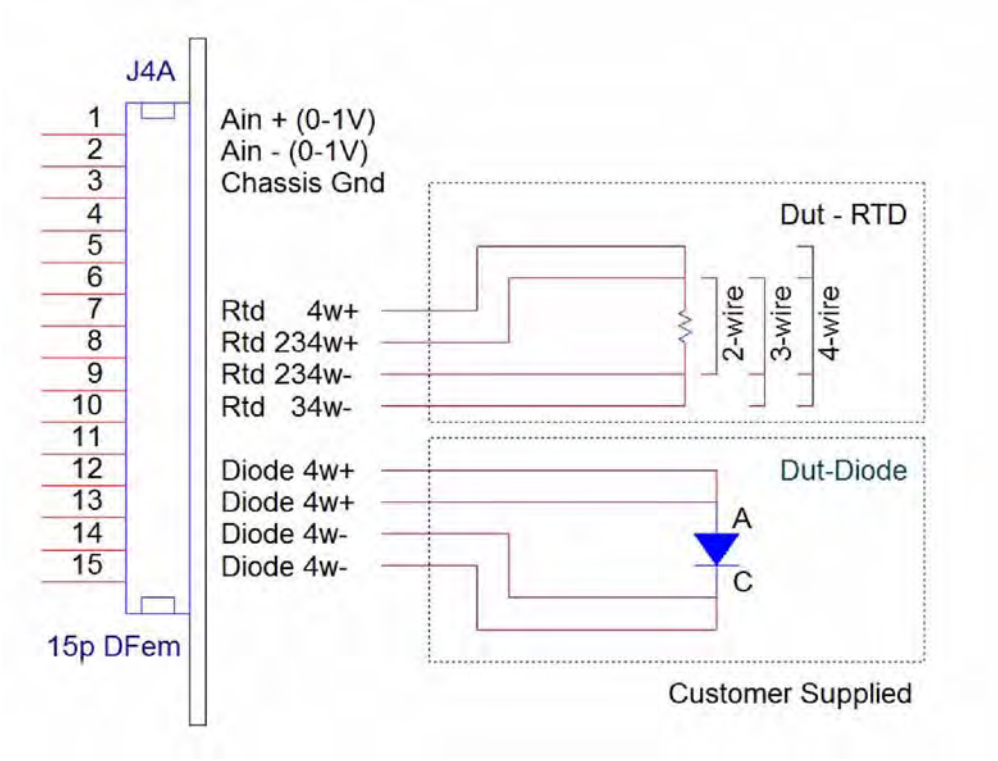
DUT Interface Pinouts

This diagram provides the pinouts for the

A Diode can be used as the DUT Sensor. The For accurate temperature readings, the diode ideality must be close to 1.0, anything over 1.25 may not be accurate.

Setup parameters F52 and F53 must be configured to match the diode being used.

Refer to DUT pin outs diagram below.



ATTENTION

FOR DIODE SENSORS

For best results, the ideality must be close to 1.0.

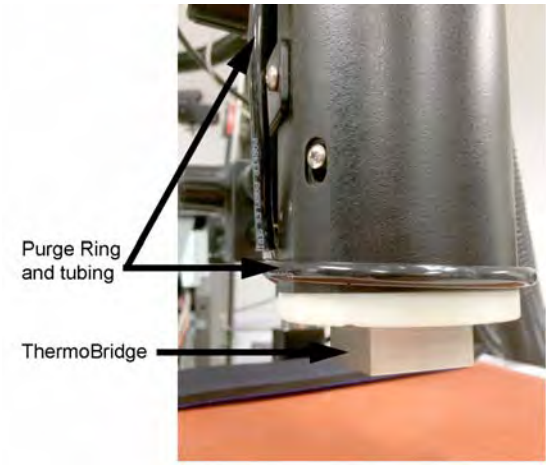
Section H: Using the Purge Collar

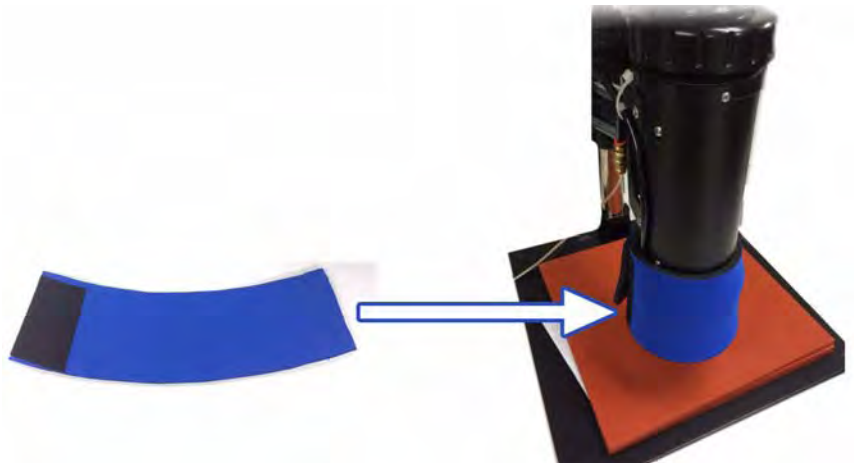
Purge Collar Overview

Introduction

The Purge Collar is designed to wrap around the thermal head to enclose and insulate the head, (optional) purge ring, and test area. The “enclosure” helps to concentrate the dry air purge and maintian a mositure-free environment around the test setup.

Connecting the Purge Collar

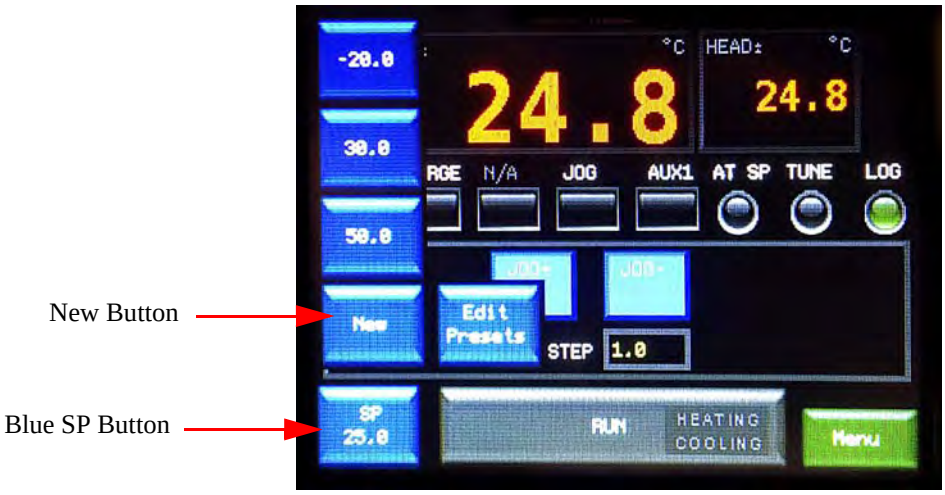




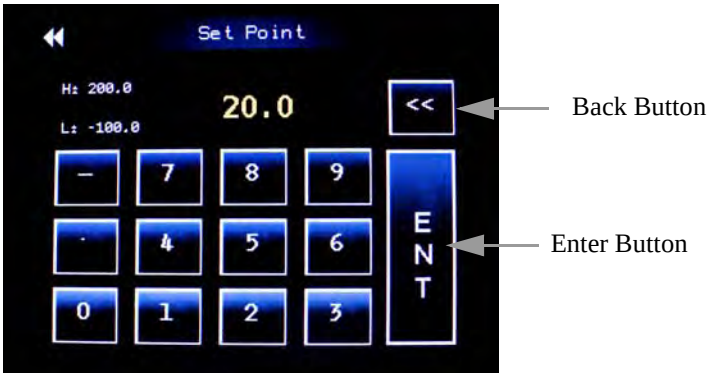
1. Return the system to ambient temperatures. Make sure the thermal head and Thermo-Bridge are at safe temperatures before starting.
2. Wrap the Purge Collar around the thermal head (optional) purge ring, and test setup. Secure with the velcro.

   If needed, use cable ties to secure the purge air lines to allow the Purge Collar to fit snug over the thermal head. Each setup is unique, ideally the collar will create an enclosure around the head and test setup to allows the purge air to circulate within the enclosure.

How To Run the System to Temperature

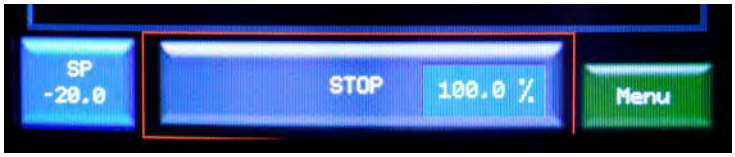


How to Run to Temperature

Step	Action
1	Visually check and verify the integrity of all power and purge air connections.
2	On the touch-screen, press the Blue SP Button. This will open the set point fly-out menu.
3	In the set point fly-out menu, press the NEW button. This will open the set point dialogue box.
4	In the set point dialogue box, enter the set point. To enter the set point: • Touch the Back Button to delete the current set point • Touch the keypad to set the new set point • Touch the ENTER button to enter the new set point. 

Step	Action
5	Press RUN and then CONFIRM RUN to run the system to the set point. The System will now cool or heat to control at the set point you have entered.

How to Stop Cooling or Heating

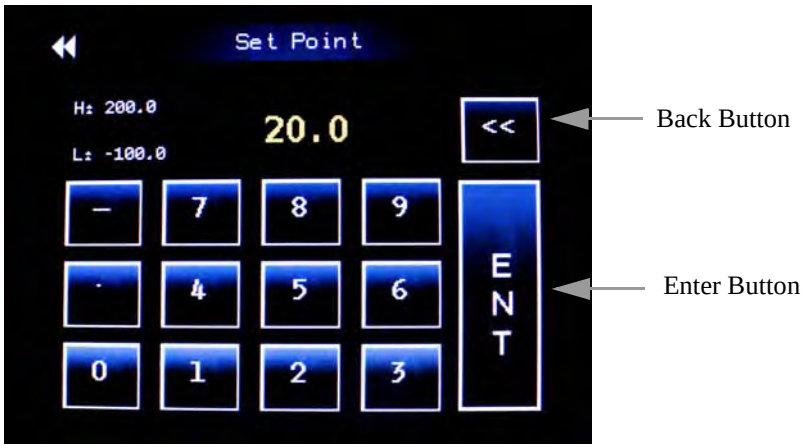
Step	Action
6	Once the system is controlling at a set point, you can stop by pressing the STOP and then CONFIRM STOP buttons. 

How To Select and Configure Set Point Presets

To Select & Configure the Presets

Three temperature set point presets are available. To select and configure the presets:

Step	Action
1	The (blue) SP button will open the Set Point flyout menu. This flyout menu contains 3 **Set Point Preset** and an **Edit Presets** button. 3 **Set Point Presets** **Edit Presets** button
2	Pressing one of the **3 Set Point Presets** will load the set point as the active set point. The active set point is the temperature the system will transition to once the RUN button is pressed.
3	To edit the presets, press the (blue) **Edit Presets** button. This will access the Edit Presets menu. Touch the temperature value you want to change. Touch to select After you touch the set point, the set point dialogue box will appear.

Step	Action
4	In the set point dialogue box, enter the new set point. To enter the set point: • Touch the Back Button to delete the current set point • Touch the keypad to set the new set point • Touch the ENTER button to enter the new set point. 
5	The Set Point Preset is now be changed.

How To Use JOG+, JOG- to Adjust Set Point

Using JOG+, JOG-

Use the JOG+ and JOG- buttons to quickly and easily change set point on the fly. The JOG buttons are typically used when the thermal head is in contact with the DUT and the end user wants to adjust the temperature.

Step	Action
1	Use the rectangular JOG soft button to activate the JOG+ and JOG- buttons. The rectangular soft button will illuminate green when the JOG buttons are active.
2	Touch the JOG STEP window to set the amount (degrees) that pressing JOG+/- will change the setpoint.
3	Press JOG+ to increase the temperature, JOG- to decrease the temperature.

Section J: Touch-Screen Controller, Advanced Operations

How To Use Program Mode

Introduction

The Program Mode allows operators to create and run temperature cycling programs. Up to 8 programs with 100 steps in each program can be entered and stored for recall and execution.

Configure Program Steps

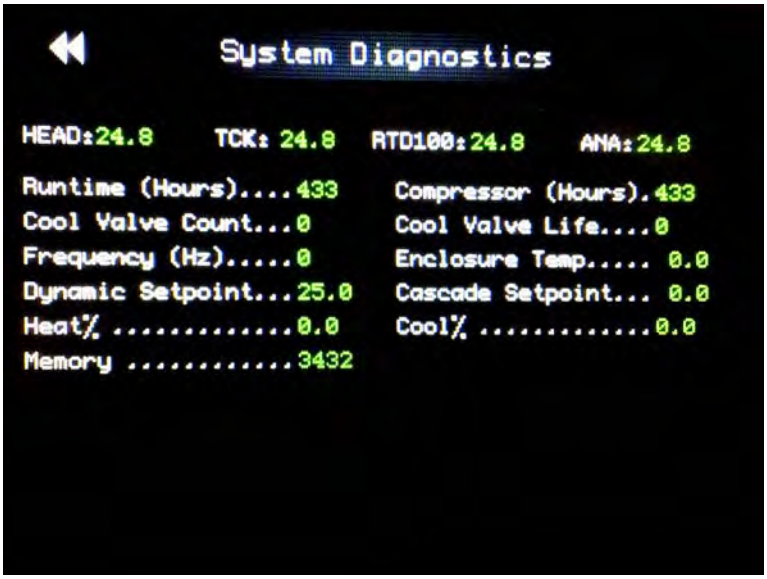
The Edit Program screen allows the operator to configure each step of the program. Up to 100 steps may be configured (step0 thru step99).Use the UP (▲) and DOWN (▼) arrows at the bottom of the screen to cycle through the steps.

Button / Field	Description
Step Type	Press to cycle between step types, the current setting will display in yellow, all other settings will displays in red. Choosing Step Type HEAD, LOOP, or AUX: **HEAD**- a head step or a “temperature step” drives the system to the set point; the following parameters may be set in a probe step: • set point- press to access the numeric keypad and enter the desired set point • Ramp Time (HH:MM:) - Set the time it will take to transition to the set point The slowest Ramp Time available is “99:59” (99hours and 59minutes). The fastest Ramp Time available is “99:99”. The system will transition as fast as possible (99:99 does NOT mean 99hours and 99minutes). • Hold Time (HH:MM:) Sets the amount of time the system will stay at the set point before advancing to the next step of the program. A countdown indicator will appear above the MENU button as the Hold Time elapses. • Next Step - a value between 0 and 100 may be entered 0-99 will advanced the program to that step. **100 is the stop command. Entering 100 as the Next Step will end the program.**

Button / Field	Description
Step Type (cont'd)	**Loop** - a loop step allows the program to "loop" or cycle through steps in a Program Loop. The following parameters may be set in a loop step: • Loop (x1)- specifies the number of loops. Value must be in the 0 - 999 range. • Loop (x1000) - specifies the number of thousands of loops (k-loops). Value must be in the 0 - 999 range. The sum of Loop (x1) and Loop (x1000) is the total number of loops in the step. The maximum number of loops is 999, 999. • N/A - this field Not Available in a loop step. • Starting Step - specifies what step the loop will start on. NOTE: Setting the starting step to 100 will stop the loop. Setting the starting step to the current step will result in an error and the loop will fail. If Step3 is a Loop step, do not enter "3" as the starting step. Example: A loop count of 5 will cause the program to execute the loop 5 times. If the program loop refers to a previously executed step as the loop start step, then that step will be executed once originally, plus the 4 more times for a total of 5 loops. When looping, a counter will appear at the top of the display screen. This counter indicates the number of loops completed. *example:* The loop counter indicating "2/ 5" means that 2 loops have been completed of 5, the third loop is in process. Once the loop has finished, the Program will advance to the next Step. It is common practice to set the step following the loop step as a Probe step that returns the system to Ambient temperatures and then shuts off the program (i.e., the Next Step value is set to 100). **AUX** - there are (3) AUX steps available (AUX0, AUX1, AUX2). These AUX steps are used to turn on or off an external device. • ON/OFF - set to turn on or off the external device with a +24VDC signal. • Next Step - Next Step - a value between 0 and 100 may be entered 0-99 will advance the program to that step. 100 is the stop command. Entering 100 as the Next Step will end the program.

Button / Field	Description
MORE button	Press to access the MORE flyout menu: • **Open Program** - press to open the Select Program dialog box. In the Select Program dialog box, Press the Program you want to load. • **Save As...** - press to open the Save Program dialog box. In the Save Program dialog box, select the Program slot you want to save your current program to. Note: this will overwrite the Program with your current program. • **Clear All** - press to clear all steps (0 thru 99) and return them to the default values. The default values are: Step Type: Probe Set Point: 30.0 Ramp Time (HH:MM) : 00:00 Hold Time (HH:MM) : 00:00 Next Step: <current step> +1 • **Clear Step:** Clears all values of the current step and replaces them with default values. • **Delete Step** - deletes the current step. This will delete the current step but add a new step 99 to the program. The new Step 99 will have default values for each parameter. • **Insert Step** - inserts a step after the currently selected step. Inserting a step will delete step 99 from the program. • **Less** - Press to exit the flyout menu.

Setup Mode: System Diagnostics Screen



Description

The System Diagnostics screen provides controller and system diagnostics including run times, voltages, operating temperatures, and current heating/cooling performance levels.

HEAD: displays current temperature of the thermal head

TCK: thermocouple K, displays current temperature of the y(ellow) K type thermocouple

RTD100: displays the current temperature of the 100ohm RTD in the thermal head

ANA: displays the current temperature of the analog temperature input

Runtime (Hours): the number of hours the system has been in operation

Compressor (Hours): the number of hours the refrigeration system (compressor) has been in operation.

Cool Valve Count: displays the number of times the currently installed refrigeration valve(s) have been activated - this value is reset when valve(s) are changed.

Cool Valve Count (Life Time): displays the number of times the refrigeration valve(s) have been activated over the lifetime of the system.

Frequency (Hz): displays the frequency of the incoming line voltage

Enclosure Temperature: displays the current temperature in the system's chassis. Excessive heat in the chassis can indicate a problem and adversely affect cooling capacity.

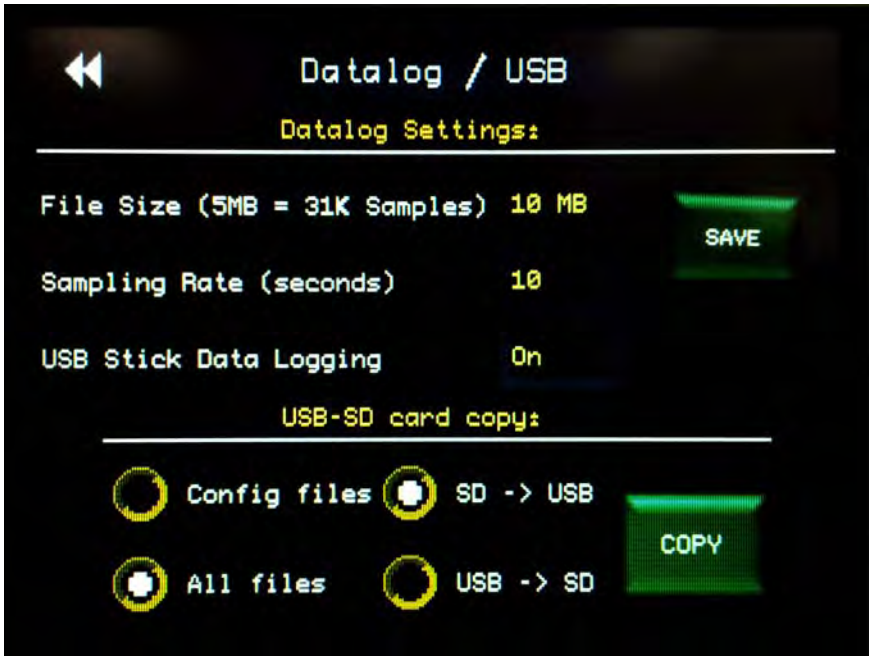
Dynamic Setpoint: used for inTEST Thermal service and controls diagnostics, displays the set point the system is using to reach and control the head temperature.

Cascade Setpoint: used for inTEST Thermal service and controls diagnostics, displays the set point the system is using to reach and control the DUT K.

Heat / Cool Percentage: displays the how much of the system's cooling or heating capacity is being used.

Memory: used for inTEST Thermal service diagnostics

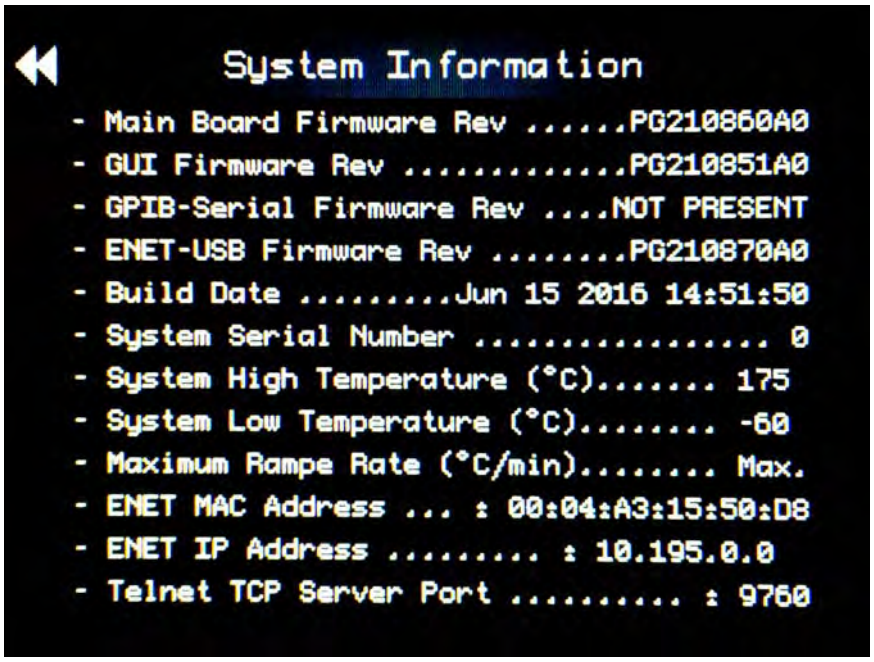
Setup Mode: Datalog / USB Screen



Description

Datalog / Stop Start	
Button / Indicator	**Description**
File Size	Sets the maximum file size and number of samples
Sampling Rate (sec)	Sets the rate (in seconds) at which the data will be sampled & logged.
USB Stick Data Logging	Set to On or Off. When set to On, data is logged to both the internal SD card and USB stick. When set to Off, data is logged only to the internal SD card.
SAVE button	Press to save Datalog Settings (File Size, Sampling Rate, USB Stick Data Logging). SAVE must be pressed when any changes are made to the Datalog Settings.
USB - SD card copy	
Button / Indicator	**Description**
Config files	Copies only the Setup Parameters (F#'s). This is typically used when copying setup files from one system to another.
All files	Copies the Log file and Setup Parameters. The log file contains all critical performance logs including temperature performance, errors, etc.
SD-> USB	Select and press COPY to copy file from the controller's SD memory card to a USB device.
USB -> SD	Select and press COPY to copy file from the USB device to the controller's SD memory card. NOTE: Diagnostics files and data log files can NOT be copied from USB to SD. Only Program and Setup files can be copied from USB to SD.
COPY	Press COPY to move file as selected (either SD to USB or USB to SD)

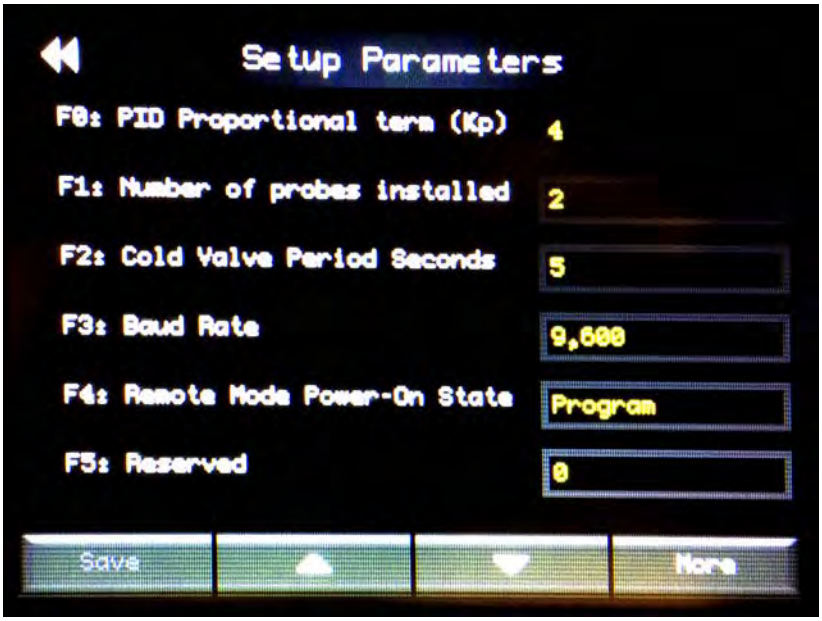
Setup Mode: System Information Screen



Description

The System Information screen provides Firmware, Serial Number, Temperature Limit, and Communication address information.

Setup Mode: Setup Parameters



Description

There are 50+ parameter used to configure system performance. These parameters control a wide range of system behaviors including temperature control, system temperature limits, remote communications setup and configuration, sensor configurations, etc.

Many of the parameters should only be changed by a qualified technician. System performance can be adversely affected by setting wrong parameter values.

The table below provides a description with the default and acceptable values for each parameter.

NOTE: You must press SAVE to apply any changes to the parameters. Pressing the back button without saving will leave the parameter unchanged. The setup parameters are stored in non-volatile memory that is retained even during power down.

Restore Defaults

Setup Parameters can be restored to their factory default settings by using the Factory Defaults button. Navigate to MENU > Go SETUP > Factory Defaults.
Press YES to restore defaults.

List of Setup Parameters

F #	Parameter Purpose	Permitted Values	Default Setting	Description
F0	PID Proportional term (Kp)	1-255	5	**For Advanced Users Only** Adjusts the temperature control algorithm - sets the value for the PID proportional term. Use a value of 12 if no ThermoBridge is installed.
F1	Number of probes installed	1-2	2	Refers to using the Head and DUT temperature sensor as the main temperature sensor. Selects the number of probes (temperature sensors) being used. Probe 1 is the Thermal Head RTD, probe 2 is the K-Type thermocouple used in the ThermoBridge or an external thermocouple connected at the system's rear I/O panel. The system is typically operated in a 2 Probe configuration.
F2	Cold Valve Period Seconds	1-10	5	**For Advanced Users Only** Sets the period for the cool duty cycle.
F3	Baud Rate	1200 2400 4800 9600 19,200 38,400 57,600 115,200	9600	For remote communications. Sets Baud Rate for USB or RS232 Serial remote communications.
F4	Remote Mode Power-On State	Immediate or Program	Program	For remote communications. Sets system's response time to remote commands: commands may be executed immediately or queued in the order received
F5	Reserved			
F6	Reserved			
F7	Tune Mode	N/A	N/A	Feature not implemented in current generation systems.
F8	RS232 Echo	Enabled - Disabled	Enabled	For remote communications. Sets up command echoing.
F9	GPIB Address	1 - 30	1	For remote communications. Sets the address for GPIB-IEEE remote communications.
F10	PID Integral term (Ki)	0 - 2000	128	**For Advanced Users Only** Adjusts the temperature control algorithm - sets the value for the PID integral term. Use a value of 512 if no ThermoBridge is installed.

F #	Parameter Purpose	Permitted Values	Default Setting	Description
F11	PID Differential term (Kd)	0 - 8000	0	**For Advanced Users Only** Adjusts the temperature control algorithm - sets the value for the PID differential term. Use a value of 0 if no ThermoBridge is installed.
F12	Reserved			
F13	Reserved			
F14	Autostart	On - Off	Off	Turns on/off the controller's autostart feature. Autostart allows the controller to remember and resume operations after a shut down.
F15	Reserved			
F16	Reserved			
F17	Probe1 Uncorrected 1	-60 to 175	0	**For Advanced Users Only** Use F17 thru F20 to apply a slope intercept curve to the temperatures that the controller is reading from Probe 1.
F18	Probe1 Corrected1	-60 to 175	0	
F19	Probe1 Uncorrected2	-60 to 175	100	
F20	Probe1 Corrected2	-60 to 175	100	
F21	Probe2 Uncorrected1	-60 to 175	0	**For Advanced Users Only** Use F21 thru F24 to apply a slope intercept curve to the temperatures that the controller is reading from Probe 2.
F22	Probe2 Corrected1	-60 to 175	0	
F23	Probe2 Uncorrected2	-60 to 175	100	
F24	Probe2 Corrected2	-60 to 175	100	
F25	ThermoSpot Low Limit	-60 to 175	-60.0	Sets the low temp. limit of the system.
F26	ThermoSpot High Limit	-60 to 175	200.0	Sets the high temp. limit of the system.
F27	DUT Low Lim.	-60°C to 175	As required Default = -50.0	Sets the low temp. limit of the DUT as read by the DUT sensor (DUTK, DUTR, DUTD, or DUTA).
F28	DUT High Lim.	-60 to 175°C	As required Default = 200.0	Sets the high temp. limit of the DUT as read by the DUT sensor (DUTK, DUTR, DUTD, or DUTA).
F29	DUT Low Diff Limit	1.0 to 300.0	50	Device Under Test Lower Operating Temperature Differential Limit)
F30	DUT High Diff Limit	1.0 to 300.0	50	Sets the maximum difference allowed between the HEAD and DUT temperatures.
F31	Settling Band C	.1°C - 10	.8	The range above or below set point that the system is considered at temperature. Sometimes referred to as the set point window.
F32	Settling Time Seconds	0 - 59	15	The amount of time the system must be within the settling band to be considered "at temperature".
F33	Minimum Ramp, °C/min	0 to 500	0.0	Sets a minimum ramp rate requirement (or temperature transition time) required by the system. Failure to ramp fast enough to stay above the minimum will result in a warning message on the controller. A setting of "0" disables the requrement for a ramp rate allowing the system to transision to temperature as fast a possible without causing an warning message.

F #	Parameter Purpose	Permitted Values	Default Setting	Description
F34	Thermal Mass	1 - 1000	32	**For Advanced Users Only** Fine-tunes thermal control to size, mass of ThermoBridge.
F35	Second Probe Type	DUTK, DUTR, DUTA, DUTD, DUTA	DUTK	Use this parameter, F35, to identify what type of probe is being used as the second probe. When F1 (Number of Probes) is set to 2 and F40 (Temperature Control Mode) is set to DUT.
F36	Reserved			
F37	Default Ramp Rate C/min	0 - 500	0	Sets the default ramp rate for all temperature transitions. a value of “0” allows the system to ramp as fast as possible and no warning will be triggered. A value of “500” will also allow the system to ramp as fast as possible but if 500 degrees per minute is not met, a warning message will trigger.
F38	Purge Set Point	-20.0 to 50.0	35	Sets the temperature threshold for the PROBE pourge air. When the system is below this temperature, the PROBE purge air will turn on.
F39	Clear at Set Point	0 (off) 1 (on)	0	**For Advanced Users Only** Clears the integrator in the PID loop.
F40	Temperature Control Mode	DUT or Normal	DUT	Normal means control off the HEAD temperature sensor. DUT means control off a DUT sensor.
F41	IP Address	xxx.xxx.xxx.xxx.	0.0.0.0	For remote communications, static IP only
F42	IP Subnet Mask	xxx.xxx.xxx.xxx.		For remote communications, static IP only
F43	IP Default Gateway	xxx.xxx.xxx.xxx.		For remote communications, static IP only
F44	GPIB Timeout, Seconds	0-120	3	For remote communications, sets GPIP timeout.
F45	Analog In Low Temp.	-250 to 500	0	Sets the Zero volt temperature of the analog (DUTA) input.
F46	Analog In High Temp.	-250 to 500	100	Sets the One volt temperature of the analog (DUTA) input.
F47	Temperature Units	Celcius - Fahrenheit	Celcius	Sets the temperature units the controller will display temperature.
F48	DUT Prop. term (Kp)	0.1 to 64.0	6.0	**For Advanced Users Only** Adjusts the DUT temperature control algorithm - sets the value for the PID porportional term
F49	DUT Integral. term (Ki)	0.1 to 64.0	0.5	**For Advanced Users Only** Adjusts the DUTtemperature control algorithm - sets the value for the PID integral term.
F50	Reserved			
F51	Reserved			
F52	Diode Ideality Factor Corr.	0.0 to 1.250	1.000	Required setup for using a DUTD (diode). Check you diode specifications for the ideality value and enter the ideality here. NOTE: Ideality must be close to 1.0 for accurate temperature readings. Any diode with an ideality value beyond 1.5 may be inaccurate.
F53	Diode Resistance Ohms	0.0 to 5.0	0.000	Required setup for using a DUTD (diode). Check you diode specifications for the resistance value.

Remote Mode

3

Chapter Overview

Introduction

This Chapter details the Remote Communications operation of the system.

In this Chapter

This Chapter is divided into the following Sections:

SECTION	See Page
Bus & User Interfaces	page 3-2
Remote Immediate (SI) & Remote Program (SP)	page 3-7
Command Summary & Programming Syntax	page 3-8
Controller Environment Information Queries	page 3-16
Controller Information Queries & Commands	page 3-18
Controller Setup Commands	page 3-24
System Operation Commands	page 3-33
Error and Status Reporting	page 3-45
Reset Commands	page 3-56

Section A: Bus & User Interfaces

Section Overview

Introduction

Remote Mode allows for control from a remote computer terminal via RS-232 or IEEE-488 GPIB. The IEEE-488 interface implementation is a TALKER/LISTENER with serial poll. Extended addressing and parallel poll capabilities are not supported. The RS-232 interface is fully configurable for baud rate.

In This Section

The following topics are covered in this Section:

Topic	See Page
Bus & User Interface	page 3-3
RS-232 Interface (optional)	page 3-3
IEEE-488 GPIB Interface (optional)	page 3-3
	page 3-6
Displaying Set Point Temperature (RS232, IEEE)	page 3-6

Bus & User Interface

Remote Mode provides for control from a remote computer or terminal using either standard RS-232 serial communication or Instrumentation Standard IEEE-488 (GPIB) communication or Ethernet. Selection of the type of interface is made automatically by the controller when it senses data on either port. Ethernet must be set up by a network administrator.
NOTE: The IEEE-488.2 requirement for lockout of front panel control during IEEE-488 control is enable and disabled with the Front Panel Lock Off (FLF) and Front Panel Lock On (FLO) commands.
In Remote Mode, the system is controlled or programmed externally by sending, over the bus, a 2 or 3 character ASCII command, often followed by arguments and ending with a terminator. See page 3-12 for Programming Syntax.

RS-232 Interface (optional)

The controller may communicate remotely through a RS-232 interface. You do not need to choose the RS-232 interface. If you send data to the RS-232 port, the controller will communicate with that port automatically. There is no longer a setup parameter that must be set to select the port for use. The RS-232 port always uses 8 data bits, no parity, and 1 stop bit. The proper baud rate must be selected using setup parameter F3. To maintain compliance with EMI emission standards, only shielded cables may be used for connection to this port. A female DB-25 connector is provided on the rear panel. The interface is set up as a DCE with DTR and CTS set always true.

IEEE-488 GPIB Interface (optional)

The IEEE-488 interface is implemented in the controller as a Talker/Listener without extended addressing or parallel poll capability. The controller will use this bus to communicate if it detects data sent to it. The primary address is set using Setup parameter F9. A standard IEEE-488 connector is provided on the rear panel. To maintain compliance with EMI emission standards, only shielded cables may be used for connection to this port.

Ethernet Interface

Ethernet communications requires setup by a network administrator. The controller supports static IP configurations (no dynamic IP may be used). The IP address, IP Subnet Mask, and IP Default Gateway can be configured with the controller's setup parameters F41, F42, F43.

For installations requiring a static IP address, users should consult their IT departments to obtain network configuration requirements. The IT department must also exclude the static IP being used.

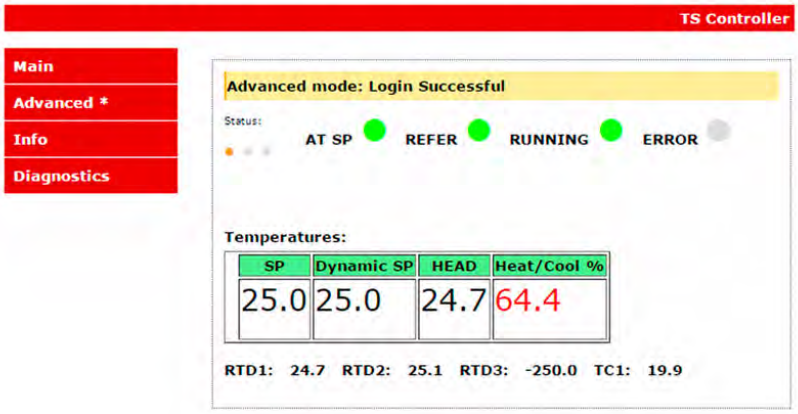
Note: Port 23 is the default port. This port should be used for control via web browser and will NOT work with telnet. Port 9760 should be used for telnet sessions.

If a telnet-based session is used, the commands, queries, and syntax are identical to the RS232 commands listed later in this chapter.

If a web browser is used, a limited web-based User Interface allows operators to monitor chiller status and change chiller set points. Accessing the User Interface will require log-in credentials. The default credentials are:

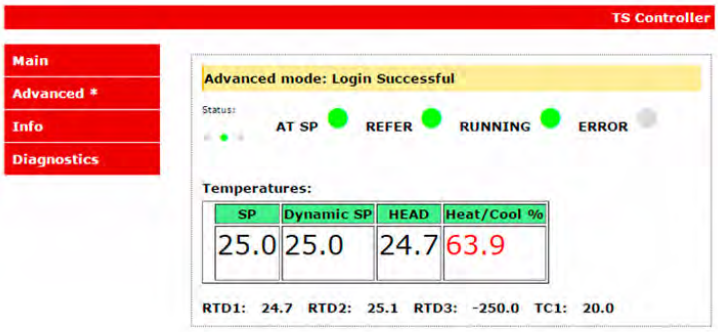
username: admin
password: iTS

Ethernet: Web Server Screens

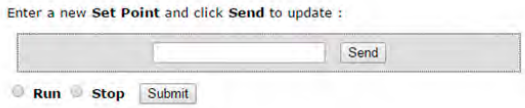


Main
displays SP (current set point) and RTD1 (current temperature)

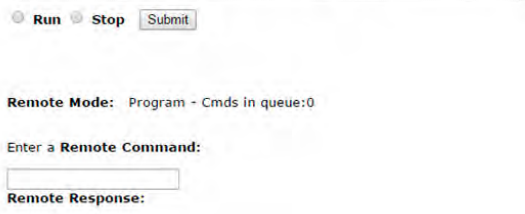
Advanced Login
Requires login.
Username: admin
Password: iTS



Advanced
monitor temperature and refrigeration system.

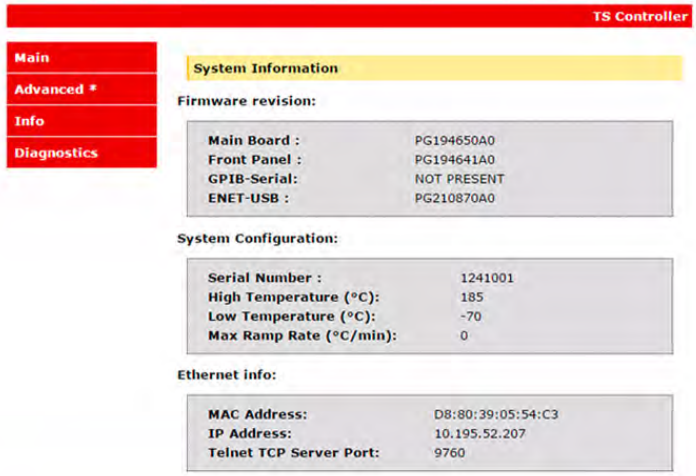


use the Set Point field to enter set points

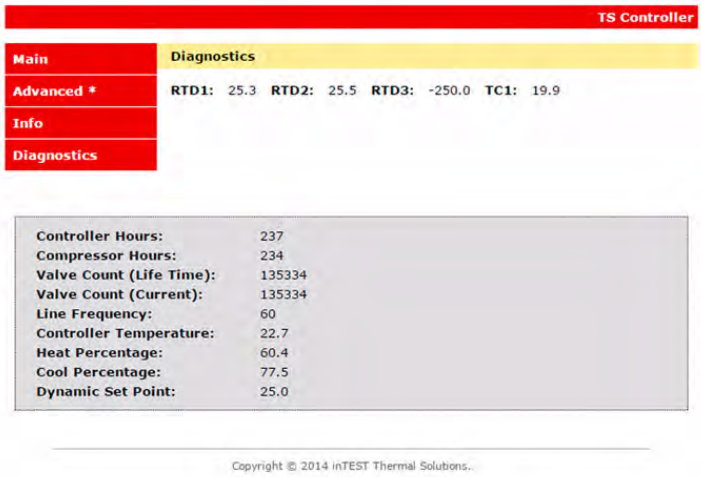


use the Remote Mode field to enter remote commands.

Ethernet: Web Server Screens



Info
diplays system info including firmware revision, system serial number, temperature limits, and network communications info.



Diagnostics
displays system diagnostics info including system runtime info, valve counts, line voltage frequency and heating/cooling performance.

Displaying Temperature (RS232, IEEE)

The temperature of Probe 1or Probe 2 may be queried. The format of the display is:

Pn TTT.T	P1 - 102.6 (Celsius)
PnF TTT.T	P1F - 102.6 (Fahrenheit)
where **n** indicates from which probe the reading originates and the **-TTT.T** indicates the numeric temperature. In single probe mode, **n** is always 1. In two probe mode, **n** can be 1 or 2. In DUT mode, **n** is 2.	

Displaying Set Point Temperature (RS232, IEEE)

The set point of Probe 1or Probe 2 may be queried. The format of the display is:

Sn TTT.T	S1 - 102.6 (Celsius)
SnF TTT.T	S1F - 102.6 (Fahrenheit)
where **n** indicates from which probe the reading originates and the **-TTT.T** indicates the numeric temperature. In single probe mode, **n** is always 1. In two probe mode, **n** can be 1 or 2. In DUT mode, **n** is 2.	

Section B: Remote Immediate (SI) & Remote Program (SP)

Introduction

The Controller can process commands either immediately upon receipt or in the sequence received.
In Remote Immediate Mode all operational commands (go to temperature, hold at temperature, etc.) and all setup commands are processed when received. Any previous command that was being processed is canceled and discarded. Query commands are also processed upon receipt, but do not cancel the underlying operational command that may be in process. This mode relies on the controlling computer to keep track of events and provide commands as needed on a real-time basis. This mode is selected by the SSI Select Immediate Mode command.

In Remote Program Mode (not related to the Program Mode that can be selected using MENU button on the controller touch screen), operational and setup commands are queued or "stacked" in a command buffer. Each command is executed to completion before the next command in the queue is retrieved and executed. Query commands are processed upon receipt and do not interrupt the processing of queued commands. This mode relies on the controller to keep track of events and can use the TE Text Echo command to notify the controlling computer when a certain place in the command queue has been reached. This mode is selected by the PN Select Active (control) Probe command.

SI (Remote Immediate Mode) or the default SP (REmote Program Mode) is typically issued at the beginning of a program sequence. However they may be issued at any point. If the controller receives a SI command and there is a command executing and/or commands in the command buffer, the executing command is interrupted and the command buffer is cleared. In either SI or SP modes, each command is discarded after execution so there is no need to clear buffer space using resets or similar commands.

Special Error Trapping Considerations for Remote Program (SP) Mode

In SP mode, queued commands sent to the controller are processed for syntax errors, and then stored in the queue for execution in sequence. It is not possible to check the command parameters at the time of receipt for errors against data that could be changed by other commands that are already in the queue, but that are not yet processed. For instance, go to temperature (GT) commands are checked to see that they request a temperature that is within the SOR (system operating range). However, if there is a SL (set DUT temperature limits) command in the queue that precedes the GT command, the SOR can change after the GT command is received, but before it is executed. Errors generated from parameter checking in SP mode will be generated at the time the command is lifted from the queue for execution. Error trapping routines will need to accommodate this error generation latency if SP mode is used. If you are having trouble determining which command is generating an error, you can try using the TE command to imbed labels in the command sequence. The error trapping routine can then record the TE generated labels that have been processed to narrow the field of discovery for chasing down the source of the error.

Section C: Command Summary & Programming Syntax

Section Overview

Introduction

In This Section

Topic	See Page
Command Summary (alphabetical)	page 3-9
Programming Syntax	page 3-12
Communication Flow Control	page 3-14

Command Summary (alphabetical)

Command Summary (alphabetical)		SP Mode Behavior	Fail-safe Mode Behavior	See Page
ASF	Autostart Off (F14)	Queued	Functional	page 3-29
ASO	Autostart On (F14)	Queued	Functional	page 3-29
CLE	Clears controller Errors and Warnings	Immediate	Functional	page 3-36
CV	Set Cold Valve Period (F2)	Immediate	Functional	page 3-30
DC	Device Clear (reset controller)	Immediate	Functional	page 3-40
DL	Delay (hold) for Interval	Queued	Ignored	page 3-36
DP	Set User Parameters to Default	Queued	Functional	page 3-30
DP1	Set Temp Reporting Decimal Places to 1	Immediate	Functional	page 3-30
DP2	Set Temp Reporting Decimal Places to 2	Immediate	Functional	page 3-43
DS	Disable SRQ (Default)	Queued	Functional	page 3-23
EF	Set Serial Echo Off	Immediate	Ignored	page 3-29
EO	Set Serial Echo On	Immediate	Ignored	page 3-29
ES	Enable SRQ	Queued	Functional	page 3-23
FLF	Front Panel Lock Off	Immediate	Functional	page 3-30
FLO	Front Panel Lock On	Immediate	Functional	page 3-30
IP	Set static IP address	Immediate	Functional	
GT	Go to C° Temperature	Queued	Ignored	page 3-34
*IDN?	Query Controller Identity	Queued	Functional	page 3-17
IP	Set static IP address	Immediate	Functional	
IPM	Set IP Address SubNet Mask	Immediate	Functional	
IPG	Set IP Default Gateway Address	Immediate	Functional	
GTF	Go to F° Temperature	Queued	Ignored	page 3-34
PF	Purge Valve Off Queued	Queued	Functional	page 3-43
PFI	Purge Valve On Immediate	Immediate	Functional	page 3-43
PN	Select Second Probe Type for Control	Queued	Functional	page 3-33
PO	Purge Valve On Queued	Queued	Functional	page 3-43
POI	Purge Valve On Immediate	Immediate	Functional	page 3-43
PT	Read Probe n Temperature	Immediate	Functional	page 3-37
PTA	Read All Probe & Set point Temperatures	Immediate	Functional	page 3-38
PTB	Read Both Probe Temperatures	Immediate	Functional	page 3-37
PTC	Read All Probe & Set point Temperatures	Immediate	Functional	page 3-38
QC	Query Last Command String	Immediate	Functional	page 3-22
QCE	Query Controller Errors	Immediate	Functional	page 3-19
QDI	Query Diagnostic Info	Immediate	Functional	page 3-16
QE	Query Error/Status String (binary	Immediate	Functional	page 3-22

Command Summary (alphabetical)		SP Mode Behavior	Fail-safe Mode Behavior	See Page
QEA	Query Error/Status String (ASCII hex)	Immediate	Functional	page 3-22
QFA	Query Setup Parameter (ASCII hex)	Immediate	Functional	page 3-28
QLP	Query Life Time Parameters	Immediate	Functional	page 3-16
QM	Query Mode Immediate or Program	Immediate	Functional	
QN	Query Controller Serial Number	Immediate	Functional	page 3-16
QP	Query Purge set point	Immediate	Functional	page 3-43
QPG	Query Remote Program Running State	Immediate	Functional	page 3-43
QR	Query Controller Temperature Range	Immediate	Functional	page 3-17
QRR	Query Ramp Rate	Immediate	Functional	page 3-43
QS	Query Set point & Control Probe Number	Immediate	Functional	page 3-38
QAI	Query system information	Immediate	Functional	
QSC	Query Control Set point & Probe Number	Immediate	Functional	page 3-39
QSI	Query System Information	Immediate	Functional	page 3-16
QSW	Read PURGE and AUX1 soft button status	Immediate	Functional	page 3-44
QU	Quit Controlling (return to idle state)	Queued	Ignored	page 3-39
QV	Query Firmware Version	Immediate	Functional	page 3-16
RA	Ramp to C° Temperature over Time	Queued	Ignored	page 3-34
RAF	Ramp to F° Temperature over Time	Queued	Ignored	page 3-34
RB	Request Error & Status Bytes (binary)	Immediate	Functional	page 3-19
RBA	Request Error & Status Bytes (ASCII hex)	Immediate	Functional	page 3-19
RDI	Read DUT PID I (Integral)	Immediate	Functional	
RDP	Read DUT PID D (Proportional)	Immediate	Functional	
RE	Request Error Byte (binary)	Immediate	Functional	page 3-18
REA	Request Error Byte (ASCII hex)	Immediate	Functional	page 3-18
RP	Read PID Constants	Immediate	Functional	page 3-26
RR	Ramp to C° Temperature at Rate	Queued	Ignored	page 3-35
RRF	Ramp to F° Temperature at Rate	Queued	Ignored	page 3-35
RS	Request Status Byte (binary)	Immediate	Functional	page 3-18
RST	Reset Controller	Immediate	Functional	
RSA	Request Status Byte (ASCII hex)	Immediate	Functional	page 3-18
RTM	Read GPIB Time-out	Immediate	Functional	page 3-23
SB	Set Settling Band Parameters for °C mode	Queued	Functional	page 3-27
SBF	Set Settling Band Parameters for °F mode	Queued	Functional	page 3-27
SC	Set Correction for Probe n	Queued	Functional	page 3-25
SD	Set DUT Temp. Differential Limits	Queued	Functional	page 3-27
SI	Select Immediate Execution Mode	Immediate	Functional	page 3-33
SL	Set DUT Temperature Limits	Queued	Functional	page 3-27
SP	Set Purge set point	Queued	Ignored	page 3-44
SPS	Set Purge set point	Queued	Ignored	page 3-44
SPA	Set F Parameter	Queued	Ignored	page 3-31

Command Summary (alphabetical)		SP Mode Behavior	Fail-safe Mode Behavior	See Page
SR	Set Default Ramp Rate in °C	Queued	Functional	page 3-44
SRF	Set Default Ramp Rate in °F	Queued	Functional	page 3-44
TCM	Temperature Control Mode	Queued	Functional	page 3-31
TE	Text Echo	Queued	Ignored	page 3-40
TF	Turn Aux/Power Port Off	Queued	Functional	page 3-39
TM	Thermal Mass			page 3-31
TO	Turn Aux/Power Port On	Queued	Ignored	page 3-39
TP	Temperature Probe Query	Immediate	Functional	page 3-31
UF	Purge Port Default State Off	Immediate	Functional	page 3-32
UO	Purge Port Default State On	Immediate	Functional	page 3-32
UP	Write Current Parameters to NV Memory	Queued	Functional	page 3-29
WDI	Write DUT PID I (Integral)	Immediate	Functional	
WDP	Write DUT PID I (Proportional)	Immediate	Functional	
WF	Aux Port 1 Default State Off	Immediate	Functional	page 3-32
WO	Aux Port 1 Default State On	Immediate	Functional	page 3-32
WP	Set PID Constants	Queued	Functional	page 3-26
WTM	Write GPIB Time-out	Immediate	Functional	page 3-30

Programming Syntax

Introduction

All commands are either 2 or 3 letters. Some commands require arguments (parameters) and some also allow optional arguments. All commands are sent with a line terminator sequence. Commands are separated from arguments by a “delimiter“. Likewise, multiple arguments are separated from each other by delimiters. The line terminator sequence follows the last argument (or the command if there are no arguments). No delimiter needs to precede the line terminator sequence. Acceptable delimiters are <space>, <comma>, or <tab> characters. The line terminator sequence is <CR><LF> to maintain compatibility with previous controllers. If compatibility is not an issue, either <LF> or GPIB EOI are acceptable line terminators. Colon and semicolon are also valid line terminators, but their use is discouraged to prevent command string concatenation.

Manual conventions

The following conventions are used to specify commands:

QN	Commands are shown in UPPER CASE in this font
<error byte>	Single entities that are not characters are described by < > symbols. The < > are not part of the command.
<comma>	Indicates a comma character (ASCII 44, h2C), but is only used when needed for clarity. In most instances a comma is shown in the text.
<space>	Indicates a space character (ASCII 32, h20), but is only used when needed for clarity. In most instances a space is shown in the text.
<tab>	Indicates a tab character (ASCII 09, h09), but is only used when needed for clarity. In most instances the text is shown tabbed.
[ ]	Brackets contain optional items. The brackets are not part of the command sequence.
{ }	Braces indicate repeating items. A sequence of items that repeats will be shown inside braces with ellipsis ... following.
|	Vertical bars separate mutually exclusive choices
nnn.n	Lower case characters in italics denote replaceable arguments and are always explained in the entry.
<CR>	Is a carriage return, ASCII 13, h0D.
<LF>	Is a line feed, ASCII 10, h0A

Command Parser & Terminators

The controller fault tolerant parser “fixes” incoming command strings before storing them in the command queue. The following “fixes” are applied:

Backspace characters are acted upon. The backspace character and the character immediately preceding it are discarded.

- Lower case commands are processed as upper case.
- Tabs are converted to space characters
- Commas are converted to space characters
- Sequential multiple space characters are reduced to a single space character.
- <LF> is the only terminator needed. <CR> will be echoed if received with echo enabled, but is not required and is internally ignored. Command strings may also be terminated with colon (:) or semicolon (;), but neither is recommended in normal use as they result in command string concatenation (see Programming Syntax, page 3-12).

- GPIB EOI is accepted as proper line termination
- Extra line terminators (<CR>, <LF>, Colon, Semicolon, EOI) are discarded

Because the fault tolerant parser makes these changes internally, you are free to use multiple spaces, tabs, or commas as delimiters to make your programs easier to read. If you prefer to write programs in lower case, the controller will not care.

Communication Flow Control

Introduction

Remote Mode operates in either of two sub-modes:

Remote sub-mode	Command	See page
Remote Program Mode *note - this is the default mode	SP	page 3-33
Remote Immediate Mode	SI	page 3-33

In Remote Immediate Mode all commands sent to the controller are acted on immediately. As commands are not queued, there are no issues relating to command buffering or flow control. Therefore, the balance of this section relates to flow control when the controller is operating in the Remote Program Mode.

In Remote Program Mode, some commands, such as queries, are acted upon immediately and are thus not stored in a command queue. However, all operational and some other commands are stored in an internal command buffer (queued). Each command in the queue is executed in the order in which it was placed in the queue (FIFO). Each command executes until it completes, regardless of the number of commands stacked behind it in the queue. When a command completes execution, such as a GT (go to temperature) command achieving the desired set point temperature, that command is removed from the queue and discarded and the next command in the queue is retrieved and executed.
The table in Command Summary & Programming Syntax (page 3-8) details which commands are queued in Remote Program (SP) Mode.

Command Buffers

When commands are sent to the controller in Remote mode, two buffers are used to handle the data. The first level buffer receives the commands as they are sent. The commands are stored in the buffer in fixed length records in the transmitted format. Commands that are immediately executed only pass through the buffer rather than being queued. The first level buffer holds 64 commands. Maximum length for received command strings, with all passed parameters and delimiters, is 32 bytes. If a command string longer than 32 bytes is received, the command string will be truncated at 32 bytes, then processed, most likely leading to an error condition. As internal processing time is available, the controller moves queued commands from the first level buffer to the second level buffer. In the process, the commands are converted to an internal format. Based upon a conservative average command length, the secondary buffer will hold approximately 410 commands. Based upon worst case conditions, which are nearly impossible to create, the secondary buffer will hold 240 queued commands. Thus, the minimum number of queued commands that can be stored in the controller under worst case conditions is 64 (first level) + 240 (second level) = 304.

Concatenating Commands

Commands can be concatenated into longer multiple command strings. There are reasons to avoid this practice.
First, some sending programs have odd responses to the timing of sending a lot of long data strings. Under some circumstance, even some versions of LabVIEW exhibit this behavior. However, our experience is that these issues are not seen when each command string is sent as a separate transmission. By separate transmission, we mean a string followed by CRLF in serial transmissions. As CRLF are transparent to GPIB, when using GPIB EOI should be asserted after each command string is transmitted.

The second reason for not concatenating command strings is that the parser and first level buffer routine inspect incoming commands and set error bits in the Error/Status String (see Error/Status String, page 3-51) or the Error Byte (see Error Byte, page 3-47) if there are problems. It can be very difficult to ascertain which command in a concatenated string caused an error bit to be set.

GPIB Flow Control

The GPIB interface contains its own flow control mechanism. When the buffers get full, the interface stops the flow of data to the controller. When the buffers again have room, the flow of data is resumed. If you are using the GPIB, you need make no special provision for flow control except to provide enough time in the GPIB time-out value while for the system to resume after a buffer full condition.

Serial Flow Control

The controller uses bits 1 and 2 of the Status Byte for data flow control (see Status Byte, page 3-49). The controller does not have hardware handshaking, nor does it use any type of imbedded flow control such as X-On/X-Off.

Not all serial communications require flow control. If the total number of queued commands to be sent is within certain limits, no flow control is required.

If you are sending 304 commands, or fewer, you do not need Status Byte flow control, but you will need to provide a 100 ms delay between each command, or a 1 second delay after each 60 commands (@9600 baud - use 3 second delay after 60 commands if using 38K baud). Probably the easiest way to provide timing for serial communications is to send an RS after sending each command and put up a read for the returning RS data. The query-read process will stop the data flow for sufficient time for the controller to process incoming commands and move them as required to the second level buffer. If using RS, we suggest you turn off Echo so you won't have to deal with the returning data.

Another way to create the required delay is to leave Echo on and put up a read for each command as it echoes back from the controller. The read process will satisfy the time delay requirement. If the delay is not provided, sending additional commands may result in data being lost in a first level buffer over-run.

If you are sending more than 304 queued commands, you will need to use flow control and you will also need to provide the time delays as explained in the previous paragraph.

Flow control uses bits 1 and 2 of the Status Byte. When the secondary buffer gets more than 75% full, bit 1 is set in the Status Byte. Bit 1 will stay set as long as the secondary buffer is more than 75% full. When the secondary buffer empties enough to be less than 50% full, bit 2 of the Status Byte is set. Thus, when bit 1 is set, you should soon stop sending data. When bit 2 is set, you can resume sending data.

If you wanted to constantly poll the Status Byte you could just use bit 1 for flow control. When the bit is set, you stop the flow of data. When the bit gets reset, you can resume the flow of data. However, all that polling can be a lot of work and a lot of bus traffic.

Bits 1 and 2 of the Status Byte will trigger the SRQ, if it is enabled (see ES Enable SRQ (Status Request Mode), page 3-23). The SRQ will send the Status Byte each time it is triggered. Enabling the SRQ makes flow control simple. You can send data until you receive a SRQ generated Status Byte transmission that shows bit 1 set. When this is received, stop sending data. When the buffer empties sufficiently to set bit 2, another SRQ generated Status Byte will be s ent. When you receive a SRQ generated Status Byte with bit 2 set, you can resume sending data. No polling is required.

Because the secondary buffer holds at least 240 queued commands, you can safely send up to 60 more queued commands after bit 1 of the Status Byte is set.

Section D: Controller Environment Information Queries

QDI Query Diagnostic Information

Query Diagnostic Info. Returns System Diagnostic information. For details on System Diagnostic information refer to the Diagnostics screen in Chapter 2.

QSI Query System Information

Query System Information. Returns System Diagnostic information. For details on System Diagnostic information.

QLI Query Life Time Parameters

Query system life time parameters. Life time parameters are found in the diagnostics screen. For details on System Diagnostic information.

QV Query Firmware Version

This command returns the current internal firmware release number in ASCII text in the following format:

QV nnn.nnn.nnn<CR><LF>

Where nnn is the number of each part of the release number, padded on the left with leading zeros if necessary.

Returned data example:

QV 007.003.001<CR><LF> Returning version 7.3.1

The syntax for the command is:

QV<CR><LF>

QN Query Controller Serial Number

This command returns the controller serial number in ASCII text in the following format:

QN n-nnnnn<CR><LF>

Where the first number n is the number designating the model group and the five digits nnnnn following the dash comprise the sequential part of the serial number.

Example:

QN 4-03376<CR><LF> Returning serial number 4-03376 (4-3376) 123

Note: When recording or reporting the serial number, always use all 7 characters in the string.

The syntax for the command is:

Q N<CR><LF>

Query Controller Temperature Range

This command returns the controller temperature range in ASCII text in the following format:

QR thhh-lll<CR><LF>

where

t = F (if Fahrenheit mode)
hhh = high limit
lll = low limit

Returned data examples:

For standard range controller in Celsius mode:
QR 200-100<CR><LF>
For standard range controller in Fahrenheit mode:
QR F200-100<CR><LF>

The syntax for the command is:

QR<CR><LF>

---

*IDN? Query Controller Identity

This is a IEEE488.2 standard command for querying the identity of an instrument on the GPIB. The expected response is manufacturer, model, serial number, firmware version.

Returned data example:

inTEST Thermal Solutions, TS Series Controller, 1241001 A0

The syntax for the command is:

*IDN?

Section E: Controller Information Queries & Commands

RS, RSA Request Status Byte

These commands retrieve the Status Byte. Reading the Status Byte will clear some of the bits. (see Status Byte, page 3-49).

If RS is used, the Status Byte is returned as a one byte binary value imbedded in an ASCII text string as follows:

RS<binary status byte><CR><LF>

If RSA is used, the Status Byte is returned as a two byte ASCII hex string imbedded in an ASCII text string as follows:

RSA<space><2 character ASCII hex string><CR><LF>

Returned data example for RSA:

RSA 47<CR><LF>

The syntax for the command is:

RS<CR><LF> Request 1 byte binary value

RSA<CR><LF> Request 2 byte ASCII hex value

Important note: The data returned in binary form by the RS command can be a line termination character that will be interpreted as a line terminator, rather than data if sent over the serial port. Use only RSA with serial communications.

RE, REA Request Error Byte

These commands retrieve the Error Byte. Reading the Error Byte will clear some of the bits. (see Error Byte, page 3-47).

If RE is used, the error byte is returned as a one byte binary value imbedded in an ASCII text string as follows:

RE<error byte><CR><LF>

If REA is used, the error byte is returned as a two byte ASCII hex string imbedded in an ASCII text string as follows:

REA<space><2 character ASCII hex string><CR><LF>

Returned data example for REA:

REA 40<CR><LF>

The syntax for the command is:

RE<CR><LF> Request 1 byte binary value

REA<CR><LF> Request 2 byte ASCII hex value

Important note: The data returned in binary form by the RE command can be a line termination character that will be interpreted as a line terminator, rather than data if sent over the serial port. Use only REA with serial communications.

RB, RBA Request Both Error Byte

These commands retrieve both the Error Byte and the Status Byte. The processing of this command, and the response, is exactly the same as if issuing a RE followed by RS, or REA followed by RSA. (see RB, RBA Request Both Error Byte, page 3-19).

Important note: The data returned in binary form by the RB command can be a line termination character that will be interpreted as a line terminator, rather than data if sent over the serial port. Use only RBA with serial communications.

QCE Query Controller Errors

Query Controller Errors, returns 16 bytes of data.
Bit 1 = Errors 1, Bit 2 = Errors 2, Bit 3 = Warnings 1, Bit 4 = Warnings 4

The following Errors and Warnings are for Standard and Advanced Diagnostics configurations.

Error 1 Bits:

E1_GPIB_RX_OVERFLOW 0x00000001

E1_FAIL_SAFE_ENGAGED 0x00000002

E1_NO_PROBE_CAL 0x00000004

E1_LINE_FREQ_NOT_DETECTED 0x00000008

E1_SYS_COMM_CHECKSUM 0x00000010

E1_SYS_COMM_OVERRUN 0x00000020

E1_SYS_COMM 0x00000040

E1_RTD1 0x00000080

E1_RTD2 0x00000100

E1_RTD3 0x00000200

E1_TC 0x00000400

E1_CJ 0x00000800

E1_GUI_PRODUCT_MISMATCH 0x00001000 // "Wrong Product Config, Please Power Cycle Unit" TS-TSR mismatch - TS must have 5.7 inch LCD, TSR must have 3.2 inch display

E1_EQUIPMENT_FAULT 0x00002000

E1_SYS_DATA 0x00004000

E1_FACTORY_NVM_NOT_SET 0x00008000

E1_TEMP_RUN_AWAY 0x00010000

E1_GPIB_SERIAL_MISSING 0x00020000

E1_FRONT_PANEL_MISSING 0x00040000

E1_ETH_USB_MISSING 0x00080000

E1_CONTROLLER_TEMP_LOW 0x00100000

E1_CONTROLLER_TEMP_HIGH 0x00200000

E1_SERIAL_RX_OVERFLOW 0x00400000

E1_REMOTE_SERIAL_COM 0x00800000

E1_PROGRAM_ILLEGAL_STEP 0x01000000

E1_EMO_SENSED 0x02000000

E1_TCK_CAL_BAD 0x04000000

E1_TCP_RX_OVERFLOW 0x08000000

E1_NVM_NOT_SAVED_BAD_PARAM 0x10000000

E1_PID_PARAMS_BAD 0x20000000

E1_AD_NO_DATA 0x40000000

E1_CLEAR_ERRORS_WARNINGS 0x80000000 // **** Special Case ******. Used to clear all errors and warnings!

Error 2 Bits:

E2_SERIAL_TX_OVERFLOW 0x00000001

E2_BLOWER1_FAULT 0x00000002 (Advanced Diagnostics option)

E2_BLOWER2_FAULT 0x00000004 (Advanced Diagnostics option)

E2_BLOWER3_FAULT 0x00000008 (Advanced Diagnostics option)

E2_BLOWER4_FAULT 0x00000010 (Advanced Diagnostics option)

E2_HEATER1_FAULT 0x00000020 (Advanced Diagnostics option)

E2_HEATER2_FAULT 0x00000040 (Advanced Diagnostics option)

E2_HEATER3_FAULT 0x00000080 (Advanced Diagnostics option)

E2_HEATER4_FAULT 0x00000100 (Advanced Diagnostics option)

E2_COOL_VALVE_FAULT 0x00000200 (Advanced Diagnostics option)

E2_CRYO_VALVE_FAULT 0x00000400 (Advanced Diagnostics option)

E2_REDUNDANT_VALVE_FAULT 0x00000800 (Advanced Diagnostics option)

E2_PURGE_BOX_HEATER_FAULT 0x00001000 (Advanced Diagnostics option)

E2_PURGE_BOX_FLOW_FAULT 0x00002000 (Advanced Diagnostics option)

E2_CRYO_EXHAUST_HEATER_FAULT 0x00004000 (Advanced Diagnostics option)

E2_BLOWER_FAULT 0x00008000 (Advanced Diagnostics option)

E2_PURGE_BOX_HIGH_TEMP_FAULT 0x00010000 (Advanced Diagnostics option)

E2_CRYO_BOOST_NOT_COOLING 0x00020000 (Advanced Diagnostics option)

E2_COOL_VALVE_NOT_COOLING 0x00040000 (Advanced Diagnostics option)

E2_BAD_EEPROM 0x00080000 (Advanced Diagnostics option)

E2_PURGE_BOX_TCK 0x00100000 (Advanced Diagnostics option)

E2_COOL_VALVE_TCK 0x00200000 (Advanced Diagnostics option)

E2_CRYO_VALVE_TCK 0x00400000 (Advanced Diagnostics option)

Warning 1 Bits:

W1_RESET_FROM_BROWN_OUT 0x00000001

W1_RESET_FROM_WDT 0x00000002

W1_SD_CARD_INIT 0x00000004

W1_SD_CARD_MISSING 0x00000008

W1_TOD_BAD 0x00000010

W1_FIRMWARE_NOT_UPDATED_FROM_VB_APP 0x00000020

W1_RAMP_TIMEOUT 0x00000040

W1_NVM_SET_TO_DEFAULTS 0x00000080

W1_SD_CARD_WRITE_PROTECTED 0x00000100

W1_SD_CARD_WRITE_ERROR 0x00000200

W1_USB_STICK_MISSING 0x00000400

W1_USB_STICK_INIT 0x00000800

W1_USB_STICK_WRITE_ERROR 0x00001000

W1_USB_STICK_WRITE_PROTECTED 0x00002000

W1_USB_STICK_NOT_ENOUGH_MEMORY 0x00004000

W1_NOT_RELEASED_FIRMWARE 0x00008000

W1_NVM_F_PARAMS_SET_TO_DEFAULTS 0x00010000

W1_BLOWER1_FAULT 0x00020000 (Advanced Diagnostics option)
W1_BLOWER2_FAULT 0x00040000 (Advanced Diagnostics option)
W1_BLOWER3_FAULT 0x00080000 (Advanced Diagnostics option)
W1_BLOWER4_FAULT 0x00100000 (Advanced Diagnostics option)
W1_HEATER1_FAULT 0x00200000 (Advanced Diagnostics option)
W1_HEATER2_FAULT 0x00400000 (Advanced Diagnostics option)
W1_HEATER3_FAULT 0x00800000 (Advanced Diagnostics option)
W1_HEATER4_FAULT 0x01000000 (Advanced Diagnostics option)
W1_COOL_VALVE_FAULT 0x02000000 (Advanced Diagnostics option)
W1_CRYO_VALVE_FAULT 0x04000000 (Advanced Diagnostics option)
W1_REDUNDANT_VALVE_FAULT 0x08000000 (Advanced Diagnostics option)
W1_PURGE_BOX_HEATER_FAULT 0x10000000 (Advanced Diagnostics option)
W1_PURGE_BOX_FLOW_FAULT 0x20000000 (Advanced Diagnostics option)
W1_CRYO_EXHAUST_HEATER_FAULT 0x40000000 (Advanced Diagnostics option)
W1_PURGE_BOX_HIGH_TEMP_FAULT 0x80000000 (Advanced Diagnostics option)

Warning 2 Bits:

W2_COOL_VALVE_NOT_COOLING 0x01 (Advanced Diagnostics option)
W2_CRYO_BOOST_NOT_COOLING 0x02 (Advanced Diagnostics option)

QE, Query Error/Status String

These commands return the 64 byte (512 bit) Error/Status String which includes the Error Byte and Status Byte. Each bit in the string represents an error or status condition. If the bit is set (1), the condition is true, if the bit is reset (0), the condition is false. The Error/Status String is divided into three parts. Reading the Error/Status String will clear some of the bits. See QE, Query Error/Status String, page 3-22) for a complete explanation of the Error/Status String.

The syntax for the command is:

QE<CR><LF> Request binary value
QEA<CR><LF> Request ASCII hex value

The QE command will return:

QE<a fixed length 64 byte binary string><CR><LF>.
Total return string length=68 bytes, including terminators.

The QEA command will return the 64 bytes of error/status data in 2 character per byte

ASCII hex in 4 lines of 16 error/status bytes each formatted as follows:

For each line:

QEA<space><first byte number in two ASCII digits><triple space><then 16 bytes of error/status data as ASCII hex with 1 space between bytes 0-7, three spaces between bytes 7-8, one space between bytes 8-15><CR><LF> as follows:

QEA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<CR><LF>
QEA 16 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<CR><LF>
QEA 32 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<CR><LF>
QEA 48 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00<CR><LF>

Programmer's note: Each of the 4 lines of text is terminated with a <CR><LF> pair. Programs that put up read commands that are satisfied by <CR><LF> will need to put up 4 reads to obtain all the data returned by QEA. The primary use of QEA is for debugging using a dumb terminal or terminal program. The line terminators are necessary in this use to format the screen output correctly. When this data is sent via GPIB, EOI is asserted only after the fourth line thus requiring only a single read to obtain all four lines of data.

QC Query Last Command

This command returns the last non-null command string sent to the controller. The returned string is the string that was delivered to the controller after the internal parser did any "fixing". It does not have to be a valid command. The string is ASCII text and is terminated with <CR><LF>. Null strings are discarded by the parser.

Example:

Previous command sent: QC returns:
Ra, -55 2,11<CR><LF> RA -55 2 11<CR><LF>

The syntax for the command is:

QC<CR><LF>

ES Enable SRQ (Status Request Mode)

For the IEEE-488 interface, SRQ is sent with the binary status. The status is not cleared until read with an RS command.

This command enables the SRQ function. Format of the command is:

ES<CR><LF>

SRQ is sent on the event of a change in state of some bits in the Status Byte. See §7.8.3

for details. For the RS-232 interface, SRQ is sent as the ASCII string:

SQ "binary status byte" <CR><LF>

The SRQ is a simple method for obtaining timely information from the controller about your process. Because the SRQ is event driven, continuous polling of the controller is not necessary. GPIB general purpose programming is beyond the scope of this manual. However, if you are not familiar with the workings and advantages of this powerful feature, we suggest that you obtain a GPIB tutorial and learn more.

DS Disable the SRQ

This command disables the SRQ function (default is DS - disabled). Format of the command is:

DS<CR><LF>

WTM Write Timeout Value

This command writes data to the GPIB Time-out setup parameter (default is WTM 3). The syntax of the command is:

WTM n<CR><LF> where n = time-out interval, 0 – 120 seconds

Example:

WTM 3<CR><LF>

RTM Write Timeout Value

This command reads data from the GPIB Time-out Setup Parameter The syntax of the command is:

RTM<CR><LF>

The response from this command is ASCII text as follows:

TM<space>nnn
Where nnn = 0 – 120
Note: TM = -1 after upgrading from prior firmware versions 9.2.1 and below.

Example response:

TM 3

Section F: Controller Setup Commands

Section Overview

Introduction

Each time Remote Mode is entered a new copy of the permanent setup parameter values is read into a working copy that is used during the Remote Mode session. If any of these working copy values have been modified by Remote Mode setup commands in this section, they will not be the same as the permanent setup parameter table values, unless an UP command is subsequently used to replace the permanent copy values with the temporary copy values. Upon exiting Remote Mode, including by power loss or fail-safe event, the temporary copy values are discarded.

The commands listed in this section change only the temporary working copy of the setup parameters. To save these values in the non-volatile permanent setup parameter table, use the UP command.

SC Set Probe Correction

WARNING

WARNING: Incorrect usage of this command may cause temperature control to be effected by invalid data and may result in unexpected temperature excursions that can cause harm to personnel and damage to equipment.

The SC command writes the data for either probe 1, or probe 2, to setup parameter fields F17-F20 or F21-F24 respectively. The data for all 4 parameters used to correct one probe must be sent with the command.

Software probe correction is achieved by entering four temperatures, U1, C1, U2, & C2, for each probe into the setup parameter table. U1 & U2 are the uncorrected, or displayed, temperatures at two points. C1 & C2 are the corrected, or actual, temperatures at those same two points.

Values for probe correction should be entered in the current scale, °C or °F, as set in setup parameter F16. There is no difference in the command syntax for Fahrenheit operation, so it is incumbent upon the user to use the correct values.

Note: Software probe correction is done separately for each probe. The following table shows the U1, C1, U2, C2 setup parameter assignments for each probe:

Probe Correction	Setup Parameter Assignments	
Correction Value	**Probe 1**	**Probe 2**
U1	F17	F21
C1	F18	F22
U2	F19	F23
C2	F20	F24

The syntax for the SC command is ASCII text as follows:

SCn U1 C1 U2 C2<CR><LF>
where n = number of probe to correct
U1, C1, U2, and C2 are floating point numbers to one decimal place.

Example:

SC1 0 2.3 100 101.3

In the event the correction parameters do not conform to the system limitations, the controller will discard the correction and set the appropriate bit in Byte 04 of the Error/Status String.

WP Set PID Constants

WARNING

WARNING: Incorrect usage of this command may cause temperature control to be adversely effected and may result in unexpected temperature excursions that can cause harm to personnel and damage to equipment if the appropriate system operating limits are not properly set.

The WP command writes the P, I, D & R data to setup parameter fields F0, F10, F11, & F12 respectively. The data for all 3 PID parameters used for PID control must be sent with the command. The R parameter is optional. The syntax for the WP command is ASCII text as follows:

WP p i d [r]<CR><LF>
where
p = proportional term constant
i = integral term constant
d = differential term constant

Example:

WP 6 5 5 30<CR><LF> (with optional R parameter)
WP 6 5 5<CR><LF> (without optional R parameter)

RP Read PID Constants

The RP command reads the P, I, D data from setup parameter fields F0, F10, F11, respectively. The data for all parameters used for PID control is returned by this command. The syntax for the RP command is ASCII text as follows:

RP<CR><LF>

Example:

RP<CR><LF>

The response from this command is ASCII text as follows:

RP<space>p<space>i<space>d<space>r<CR><LF>
where
p = proportional term constant
i = integral term constant
d = differential term constant

Example: RP 6 5 5 30<CR><LF>

SL Set DUT Temperature Limits

The SL command sets the lower DUT limit (setup parameter F27) and the upper DUT limit (setup parameter F28).

The command sets the lower and upper temperature limits at the same time. The syntax of the command is:

SL lll.l uuu.u<CR><LF>
where:
lll.l is the lower limit (F27)
uuu.u is the upper limit (F28)
lll.l & uuu.u are one decimal place numbers

Example:

SL -75 130<CR><LF>
Sets DUT limits to -75° & +130°

SD Set DUT Temperature Differential Limits

The SD command sets the lower DUT temperature differential limit (setup parameter F29) and the upper DUT temperature differential limit (setup parameter F30).

The command sets the lower and upper temperature differential limits at the same time. The syntax of the command is:

SD lll.l uuu.u<CR><LF>
where
lll.l is the lower differential limit (F29)
uuu.u is the upper differential limit (F30)
lll.l & uuu.u are one decimal place floating point numbers

Example:

SD 50 30<CR><LF>SD lll.l uuu.u<CR><LF>
where
lll.l is the lower differential limit (F29)
uuu.u is the upper differential limit (F30)
lll.l & uuu.u are one decimal place floating point numbers

Example:

SD 50 30<CR><LF>

SB, SBF Set Settling Band Parameters

The SB & SBF commands set the set point temperature bandwidth and minimum stable time requirements for “satisfying” a go to temperature request (setup parameters F31 & F32).

The command sets the set point temperature bandwidth and stable time at the same time. The syntax of the command is:

SB tt.t ss<CR><LF> Celsius mode
SBF tt.t ss<CR><LF> Fahrenheit mode

where
tt.t is degrees of temperature bandwidth (± from set point) allowed to “satisfy” a go to temperature request). The valid range of this parameter is 0.1°C (0.2°F) to 10°C (18°F). Default = 0.8°C

ss is seconds of stable temperature (within above bandwidth) required to “satisfy” a go to temperature request. The valid range is 0 to 59 seconds. Zero has the special meaning that the set point will be considered “satisfied” the first time the controlling probe temperature reaches or passes through the set point (± the allowable bandwidth as above). Default = 15 seconds

Example:

SB .8 15<CR><LF>
SBF 1.3 20<CR><LF>
Sets the settling band parameters to the °C mode default values of 0.8° and 15 seconds or 1.3°F and 20 seconds in Fahrenheit mode.

Note: If the second parameter is omitted, it is assumed to be 0. If the parameters passed with this command are invalid, Bit 2 of the Error Byte will be set.

QFA Query Setup Parameter Value

This command returns the current value of a setup parameter field from the temporary working copy of the setup parameter table. QFA requests the data to be returned in ASCII text form. The syntax of the command is:

QFAnn<CR><LF>
(Query for ASCII text)

Where nn is a one or two byte ASCII text decimal number of the setup parameter field to be returned.

For parameters numbered 0 through 16:

The value is returned in two bytes of ASCII hex in the format 00 in an ASCII string followed by CR/LF as follows:

QFAnn hh<CR><LF>
Where nn is the setup parameter number in 2 bytes of ASCII text decimal (leading zero if required) and hh is the ASCII hex value.

For parameters numbered 17 through 41:

The values returned for parameters numbered 17 through 41 are expressed as ASCII decimal numbers. All except F32, F40, & F41 are expressed with one decimal place. F32, F40 & F41 are expressed as integers. Negative values will have a leading minus sign. If the value is a Fahrenheit value, the character “F” will be appended to the end of the string. CR/LF terminates the string. Format of the returned data is:

QFAnn -123.4F<CR><LF>

Note: The values returned by QFA are the temporary working values. These values are originally obtained for Remote Mode from the permanent values used by Local, Program, and Setup Modes. Each time Remote Mode is entered a new copy of the permanent setup parameter values is read into a working copy that is used during the Remote Mode session. If any of these working copy values have been modified by Remote Mode setup commands (See

Controller Setup Commands Introduction on page 3-24 and UP Write Temporary Setup Parameters to Permanent Table, on page 3-29) they will not be the same as the permanent copy values unless an UP command was subsequently used to replace the permanent copy values with the temporary copy values. Upon exiting Remote Mode, including by power loss or fail-safe event, the temporary copy values are discarded.

UP Write Temporary Setup Parameters to Permanent Table

UP writes the current working copy setup parameters to the permanent setup parameter table. See Controller Setup Commands Introduction on page 3-24 for explanation. This command uses no arguments. The command writes all parameters at one time. The syntax of the command is:

UP<CR><LF>

There is no response from the controller to this command.

AF, AO Auto Start Off On

The AO and AF commands set Auto Start on or off. The Auto Start allows the controller to remember and resume system status after a power down. Command Syntax:

ASF<CR><LF>

ASO<CR><LF>

EO, EF Serial Communications Echo ON, OFF

These commands turn serial communications echo on or off by changing the temporary setup parameter F8. The permanent setup parameter table is not loaded with resultant F8 value unless an UP command is used. The syntax of the commands is:

EO<CR><LF> Set temporary F8=1, turn echo ON
EF<CR><LF> Set temporary F8=0, turn echo OFF

EO and EF are immediate commands in Remote Program mode.

RTM Read Timeout value

The RTM command changes the GPIB Bus time-out interval. It has a range of 0 to 120 seconds. Zero is no time out. This value is the amount of time in seconds that the controller will wait for a response. Two different conditions can result in a bus time-out. 1) If the interface could not accept a data byte that was written (interface is not a talker) and 2) if a response was waiting to be sent and the host never requested the data within the GPIB time-out setting. Case 1 will cause error status bit 32:1(0x02) to be set. Case 2 will cause error status bit 32:2(0x04) to be set. Error status bit 37:3 (0x08) is set for transient Fail-Safe conditions. This is different from bit 00:3 which is only set if the fail-safe is constant. This is to help detect noise on the power source since that can cause bus-failures too. Command Syntax is:

RTM <CR><LF>

The ASCII Command response format is:

TM nnn<CR><LF>
where nnn is time-out value in seconds. -1 or 0 = no time-out. -1 will only occur if the controller has been upgraded from prior versions that did not have this parameter.

WTM Write Timeout value

WTM command allows GPIB bus time-out to be set. Command Syntax is:

WTM nnn<CR><LF>
Where nnn is the new timeout value to be used.
This command does not have a response.

AF, AFO Auto Start Off, On

AF and AO sets the Auto Start condition (parameter F14). Set to “On”, the controller will recall status after a power down event.

FLF, FLO Front Panel Lock Out Off, On

FLF and FLO sets the front panel lockout state during remote operations. Set to “On”, the controller front panel will be disabled and display a remote lock icon.

CV Set Cold Valve Period

Set the cold valve period with an integer from 1 to 10 (seconds). This command changes the status of parameter F2. It is used to limit the number of cycles placed on the cold valve.

DP Set User Parameters to Default

Sets all user parameters (i.e. F#s) to their factory default values.

SPA Set F Parameter

Use to change setup parameters. SPA gives remote users access to every F parameter in the setup parameters. Set F parameter *n* to a value *x*. *X* can have a sign and decimal point depending on the F parameter being set. For example:

SPA<sp>1<sp>2<cr> would set F1 (Number of probes) to a value of 2.

or

SPA<sp>35<sp>1<cr> would set F35 (second probe type) to DUTK (1=DUTK, 2=DUTR, 3=DUTD, 4=DUTA)

TCM Temperature Control Mode

Changes the temperature control mode between DUT (Device Under Test) mode and Normal mode. 1= DUT, 0= Normal.

Normal controls and reads temperature based on the HEAD temperature. This setting relies on the RTD sensor located in the thermal head.

DUT controls and reads temperature based on a second sensor (DUTK, DUTR, DUTD, DUTA). DUTK is available as either the sensor at the ThermoBridge or as a rear I/O port sensor.

TM Thermal Mass

Sets the thermal mass. Valid values are 1-1000. This command changes parameter F34.

TP Read Probe Temp

Read Probe Temp. TPn<CR><LF>

Read Probe Temp. n = 1-4.

1 = DUTK

2 = DUTR

3 = DUTD

4 = DUTA

UF, UO Purge Port Default State Off, On

Sets the default state of the Purge Port. Syntax:

UF<CR><LF>

UO<CR><LF>

Section G: System Operation Commands

SSI Select Immediate Mode

This command selects the immediate mode of operation. In immediate mode, a command is executed immediately when it is received over the interface. If a previous command is executing, it is canceled. Query or status commands do not cancel executing operating commands. Refer to Command Summary & Programming Syntax, page 3-8. Commands shown as “immediate” in that table are executed without interrupting the currently executing command. Command syntax is:

SI<CR><LF>

The controller does not respond when receiving this command.

Note that if SI is sent while the controller is in SP mode, all SP mode queued commands are discarded upon receipt of the SI command.

PN Select Active (control) Probe

This command selects the second probe type (F35). Choose between DUTK, DUTR, DUTD, DUTA

1= DUTK

2= DUTR

3=DUTD

4=DUTA

The default when entering the Remote Mode of operation is DUTK (1.)

Command syntax is:

PN n<CR><LF> Where n is probe number to use for control

Example:

PN 1<CR><LF> Selects DUTK (default) for control.

GT, GTF Go To Temperature

This command instructs the controller to control to the set point specified in the command. The arguments for the command include a set point temperature, and optionally, a ramp time-out value. Bit 4 (set point reached) of the Status Byte is set when the system has stabilized within the settling band as defined by setup parameters F31 and F32. The default settling requirement is within 0.8°C of the set point for 15 seconds. The system will attempt to reach the specified set point as quickly as possible. The command syntax is:

GT<space>[-]ttt.t [rrr.r]<CR><LF> Celsius mode
GTF<space>[-]ttt.t [rrr.r]<CR><LF> Fahrenheit mode

Where
ttt.t is the final set point temperature
rrr.r is the minimum ramp rate value for the Ramp Time-out Function

Examples:

GT 125<CR><LF> Celsius mode - go to 125°C
GTF 180<CR><LF> Fahrenheit mode - go to 180°F
GT 125 5<CR><LF> Go to 125°C and set warning if ramp rate falls below 5°C/minute

---

RA, RAF Ramp to Temperature

This command executes a ramping function. The arguments for the command include a ramp temperature and a ramp time, and optionally, a minimum ramp rate value. The controller will execute a linear ramp from the current temperature to the specified temperature over the specified time period. Bit 3 (interval complete) of the Status Byte is set when the system has completed the ramp. The command syntax is:

RA [-]ttt.t hh mm [rrr.r]<CR><LF> Celsius mode
RAF [-]ttt.t hh mm [rrr.r]<CR><LF> Fahrenheit mode
Where
ttt.t is the final set point temperature
hh are the hours to be used for the ramp (range is 0- 99 hrs)
mm are the minutes to be used for the ramp (range is 0-59 minutes)
rrr.r is the minimum ramp rate value for the Ramp Time-out Function in °/minute

Examples:

RA -55 2 15<CR><LF> Celsius mode
RAF -55 2 15<CR><LF> Fahrenheit mode
Ramps to -55° over 2 hours and 15 minutes

RA -55 2 15 1<CR><LF>
Ramps to -55°C over 2 hours and 15 minutes and sets minimum ramp warning if unable to maintain at least 1°C/minute ramp rate.

RA & RAF are time based commands. The controller establishes a linear ramp rate based upon the original starting temperature, the final set point temperature, and the time allowed. If the system is unable to maintain the ramp rate specified by this command, the command will terminate before the set point is reached. This issue is most commonly seen with refrigerated systems where the cooling rate decays as the temperature is reduced.

Example: Starting at 0°, an RA command is issued to go to -60° in 20 minutes. The controller sets a ramp speed of 3°/minute. In this example, the system is capable of cooling at better than 3°/minute to -40°, and about 2°/minute to -60°.

The system is capable of cooling at the specified 3°/minute to -40° so the 3°/minute ramp speed is maintained to -40°. 13.3 minutes of our allotted 20 minute ramp time has been used. Because the system in this example is only capable of cooling at 2°/minute at temperatures below -40°, in the remaining 6.7 minutes of allotted ramp time, the system temperature will only cool down another 13.4°. Thus, at the end of the specified ramp time, the temperature will be -53.4°, not the 60° specified in the command.

To make sure that the final temperature is reached, either make certain that the specified ramp rate is within the capability of the system for the entire ramped temperature range, or follow the RA command with a GT command with the same temperature set point.

RR, RRF Ramp to Temperature at Specified Rate

This command executes a ramping function. The arguments for the command include a ramp temperature and a ramp rate, and optionally, a minimum ramp rate value. The controller will execute a linear ramp from the current temperature to the specified temperature at the rate specified. Bit 3 (interval complete) of the Status Byte is set when the system has completed the ramp. The command syntax is:

RR [-]ttt.t ddd.d [rrr.r]<CR><LF> Celsius mode
RRF [-]ttt.t ddd.d [rrr.r]<CR><LF> Fahrenheit mode

Where
ttt.t is the final set point temperature
ddd.d is the ramp rate in °/minute (range is 0.1 to 999.9°/minute)
rrr.r is the minimum ramp rate value for the Ramp Time-out Function in °/minute

Examples:

RR -55 15<CR><LF> Celsius mode
RRF -55 15<CR><LF> Fahrenheit mode
Ramps to -55° at 15°/minute

RR -55 15 10<CR><LF>
Ramps to -55°C at 15°/minute and sets minimum ramp warning if unable to maintain at least 10°C/minute ramp rate.

RR & RRF are rate based commands. The controller establishes a linear ramp rate based upon the original starting temperature, the final set point temperature, and the ramp rate specified. The controller then computes the amount of time the command should require. If the system is unable to maintain the ramp rate specified by this command, the command will terminate at the end of the time allotted in the same manner as the RA and RAF commands. When the set point has been reached, to the satisfaction of the settling band, the command terminates, regardless of whether the expected ramp time has been used.

CLE (Clear Errors)

Command clears TS and TSR touch screen controller errors. If the error condition persists, corrective action may be required.

DL Delay (Dwell Interval)

The DL command causes the controller to maintain the set point temperature for a specified interval of time before executing the next command. The Interval Complete bit of the Status Byte is set when the system has completed the requested delay interval. The command syntax is:

DL hh mm<CR><LF>
Where
hh are hours to be used for the delay interval
mm are the minutes to be used for the delay interval

Example:

DL 1 20<CR><LF>
Will hold the current set point for 1 hour and 20 minutes

Note: The DL command can also be used at the beginning of a program, or after a QU command to insert a “wait” interval before the next command will execute (in Remote Program (SP) mode.

PT Read Temperature

This command is used to read the specified probe temperature. Control Set point (CSP). The format of the command is:

PT n<CR><LF>
where
n is the number (1 or 2) of the probe to be read. Probe 2 is only available if enabled by in the Setup Parameter.

Example:

PT 2<CR><LF>
reads the second probe temperature.

The response is ASCII text with temperature as a one decimal place number. Negative temperatures are preceded by a minus sign. Temperatures read in Fahrenheit mode have an “F” character inserted after the probe number:

Tn -ttt.t<CR><LF> for Celsius mode
TnF -ttt.t<CR><LF> for Fahrenheit mode
where
n is the probe number (1 or 2) and
ttt.t is the temperature.

The ttt.t value is not fixed length, but always includes one decimal place. A minus sign precedes negative values.

Response examples:

T1 102.0<CR><LF> Probe 1 reports a temperature of 102.0°C
T1F -74.3<CR><LF> Probe 1 reports a temperature of -74.3°F

PTB Read Both Probe Temperatures

This command is internally processed identically to sending PT 1 and then PT 2. The only difference in the response to this command is that if the second probe is not enabled in setup parameter F1, the command does not generate an error, or a response for probe 2. The syntax of the command is:

PTB<CR><LF>

A typical response with F1=2 is:

T1 102.6<CR><LF>
T2 130.3<CR><LF>

A typical response with F1=1 is:

T1 102.6<CR><LF>

PTA Read All Probes and User Set point

This is a legacy command that only supports reading 2 Probes (Head and DUTK).

Use the TP command to read temperatures of DUTK, DUTR, DUTD, and DUTA.

This command is internally processed identically to sending the following 3 commands:

PT 1 PT 2 QS

The response for probe 2 temperature is only returned if probe 2 is enabled by the F1 setup parameter.. Thus typical responses are:

For normal probe 1 control when two probes are enabled in F1:

T1 102.6<CR><LF>
T2 130.3<CR><LF>
S1 110.0<CR><LF>

For normal probe 1 control only one probe is enabled in F1:
T1 102.6<CR><LF> For normal probe 1 control when
S1 110.0<CR><LF> only one probe is enabled in F1

PTC Read All Probes and Set points

This command is internally processed identically to sending the following 3 commands:

PT 1 PT 2 QS QSC

The response for probe 2 temperature is only returned if probe 2 is enabled by F1=2. The response for QSC is only returned if there is a control set point in use for a controller ramp. Thus typical responses are:

For normal probe 1 control when two probes are enabled in F1:

T1 102.6<CR><LF>
T2 130.3<CR><LF>
S1 110.0<CR><LF>

For normal probe 1 control only one probe is enabled in F1:
T1 102.6<CR><LF> For normal probe 1 control when
S1 110.0<CR><LF> only one probe is enabled in F1

QS Query Set point & Control Probe Number

This command returns the current set point and control probe number as follows:

Sp [-]nnn<CR><LF> For Celsius mode
SpF [-]nnn<CR><LF> For Fahrenheit mode
Where
p is the number of the controlling probe and
nnn is the current set point with leading minus sign if needed.

Returned data example: Control with probe 1 to -85°F

S1F -85.0<CR><LF>

The syntax for the command is:

QS<CR><LF>

QSC Query Control Set point & Probe Number

This command returns the current control set point (CSP) and control probe number as follows:

Cp [-]nnn<CR><LF> For Celsius mode
CpF [-]nnn<CR><LF> For Fahrenheit mode
Where
p is the number of the controlling probe and
nnn is the current control set point with leading minus sign if needed.

The control set point is the internally generated set point that the controller uses for temperature control during controlled rate ramps (RR, RRF, RA, RAF commands).

Returned data example:

Current CSP is 85.7°F using probe 1
C1F -85.7<CR><LF>

The syntax for the command is:

QSC<CR><LF>

TO & TF Turn Aux/Power Port On

The TO command turns the Auxiliary/Power port on. The TF command turns the port off. The default setting upon entering Remote Mode is determined by setup parameter F5. The current state of the Aux/Power Port is reported by Byte 35, Bit 4 of the Error Status String which can be accessed using the QE or QEA commands. See QE, Query Error/Status String, page 3-22.

The Auxiliary/Power port is normally implemented as a TTL output available at the back of the controller circuit board. The port is typically used to control an optional remote power switch.

The command syntax is:

TO<CR><LF>
Turn the Aux/Power Port on

TF<CR><LF>
Turn the Aux/Power Port off

QU Quit Controlling

The QU command stops the controller from controlling temperature. All heating and cooling is turned off, the blowers are set to the mode determined by setup parameter F15 (Blower Shut-off Mode). The refrigeration compressor port and the aux/power ports are not affected. The command syntax is:

QU<CR><LF>

After a QU command, the display will show:

U1c 39_2 (Celsius mode with current temperature display)

U1F 39_2 (Fahrenheit mode with current temperature display)

Note: The controller can also be commanded to STOP controlling the temperature by issuing a DC command or IEEE-488 standard I/O statements ABORTIO, Device Clear, or Selected Device Clear. The primary functional difference is that the QU commands simply returns the system to the “idle” state. The DC and GPIB commands Device Clear and Selected Device Clear, but not the GPIB function Interface Clear, result in a system reset.

DC Device Clear

The DC command executes the same routines as a GPIB Device Clear or Selected Device Clear. The command returns the controller nearly to its power on state with a new copy of the permanent setup parameters in the working setup parameter table. The communications and command buffers are cleared and all working copy setup parameters are lost.

The syntax of the command is:

DC<CR><LF>

TE Text Echo

TE is a queued (Remote Program Mode) command that returns the argument text when processed. The purpose of the command is to allow a text string to be imbedded in a command sequence. When the command is executed, the string supplied as the command’s argument is returned to the user. The text string can be any ASCII text up to 28 bytes in length that does not contain 0x00, or <CR> or <LF>. Spaces, commas, quotation marks, etc. are all acceptable. Note that colons and semicolons are not acceptable as they are command terminators.

Use example:

The objective is to:
go to 120°, hold for 30 minutes, then run a test that takes 10 minutes, then go to -50°, hold for 30 minutes then run a second test that takes 10 minutes.
There are two ways to accomplish the goal.

One way is to run the controller in Remote Immediate Mode (SI command) and let the controlling computer do the timing work and keep track of events in the following manner:

Send **SI**
Set controller to Remote Immediate Mode

Send **GT 120**
Go to 120°
Watch for “set point reached” (bit 4) in the Status Byte
Wait for 30 minutes
Run test (controller will continue to hold 120°)

Send **GT -50**
Go to -50°
Watch for “set point reached” (bit 4) in the Status Byte
Wait for 30 minutes
Run test (controller will continue to hold -50°)

Send **QU**
Quit controlling

The other way is to run the controller in Remote Program Mode (SP command) and let the controller do the timing work and keep track of events, notifying the controlling computer when it is time to run the tests. The sequence uses the TE command to let the controlling computer know when it is time to run the tests. Example:

Send **SP**
Set controller to Remote Program Mode

Send **GT 120**
Go to 120°

Send **DL 0 30**
Hold for 30 minutes

Send **TE Begin1**
Echo “Begin” when 30 minute hold is finished

Send **DL 0 11**
Hold for 11 minutes (10 min for test + 1 min for starting and stopping the test)

Send **GT -50**
Go to -50°

Send **DL 0 30**
Hold for 30 minutes

Send **TE Begin2**
Echo “Begin2” when 30 minute hold is finished

Send **DL 0 11**
Hold for 11 minutes (10 min for test + 1 min for starting and stopping the test)

Send **QU**
Quit controlling
Wait for “TE Begin1" then run test 1
Wait for “TE Begin2" then run test 2
Complete

Programming note: The 28 byte maximum length of the TE command text begins immediately after the TE. Internally, the controller accepts all characters up to the first terminator. When the string is processed, all leading blank characters are stripped and then one leading space is placed between the TE and string upon return.

Thus, all of the following:

TE123 -45,6<CR><LF> no leading space

TE 123 -45,6<CR><LF> one leading space

TE 123 -45,6<CR><LF> multiple leading spaces

TE 123 -45,6<CR><LF>

DP1, DP2 Set Remote Mode Temperature Decimal Places

These commands temporarily change the number of decimal places returned by PT and related temperature query commands. These commands do not change any other returned temperature data, such as set point data, nor do they affect the data displayed on the controller front panel.

The default number of decimal places for temperature data is 1 place. Each time the controller enters Remote mode, or undergoes a reset, this default value is restored. If DP2 is subsequently sent to the controller, then all temperature data will thereafter be sent with two decimal places. Sending DP1 or issuing a reset, will change the number of decimal places returned to 1.

The DP2 command does not change the accuracy of the temperature data. Its only purpose is so that users who are doing very low gradient and precision work can recognize the difference between a 0.1° temperature change and a smaller change that simply resulted in a rounding event.

PF, PO

Purge valve Off, On. This will be queued if in Remote Mode. This state is saved with the UP command.

PFI, POI

Purge valve Off, On immediately. This will be not be queued if in Remote Mode. This state is saved with the UP command.

QP Query Purge Set point

Query Purge set point

QPG Query Remote Program Running State

Query Remote Program Running set point. Queries whether or not a Remote Program is currently running. The response will be either “PGM” or

---

QRR Query Ramp Rate

Query Ramp Rate. Ramp rate is in °C/minute

SP Set Purge Set point

Set the the Purge Set Point (F38) in °C. If the system goes below this temperature, the PROBE purge air will turn on.

SR, SRF

Set the ramp rate in °C (SR) or °F (SRF)

Section H: Error and Status Reporting

Section Overview

Introduction

There are three sources for error and status information:

- The Error Byte
- The Status Byte
- The Error/Status String (64 bytes)

In This Section

Topic	See Page
Relationship of Error Byte, Status Byte, and Error/Status String	page 3-46
Error Byte	page 3-47
Status Byte	page 3-49
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Relationship of Error Byte, Status Byte, and Error/Status String

The Error/Status String is the Error Byte (not a replica) and byte 01 of the Error/Status String is the Status Byte (again, not a replica). Thus, there cannot be inconsistencies in the behavior of the separately accessible Error Byte and Status Byte as compared to their Error/Status String counterparts as they are now the same internal data.

Because the Error Byte and Status Byte can be read either directly or as part of the Error/Status String, the user is free to decide if the additional information available in the Error/Status String is needed. If the additional information is needed, then reading the Error/Status String will provide both the Error Byte and Status Byte information as well as the additional Error/Status String data in a single read. However, if the additional information is not required, the user can use the simpler single byte read commands to read the Error Byte and the Status Byte without incurring the overhead of having to parse the 64 byte Error Status String. The behavior of the Error Byte and Status Byte bits is the same regardless of the method used to read them.

TIP: When starting a new session, the easiest way to clear all the event bits in all of the error/status data is to issue a QE command.

Error Byte

Effect of the Error Byte on Bit 7 of the Status Byte

The Error Byte is a one byte block of data in which each of the eight bits reports the status of an item. Each bit can be either 0 or 1. A bit with a value of 0 is “reset” and is equivalent to logical false. A bit with a value of 1 is “set” and is equivalent to logical true.

The bits in the Error Byte report the status of the following items:

Bit	Definition - Error Byte	Type
0	Aux Input toggled (typically optional door switch open/closed)	Event
1	System Fault Alert	Event
2	Illegal argument string. A valid command was received but the accompanying argument(s) was/were not valid	Event
3	Fail-safe tripped	Event
4	Data in bytes 02-31 of the Error/Status String set	Event
5	Undefined command. A command string was received that is not recognized by the controller	State
6	Command buffer overrun. Commands were sent to the controller when the command buffer was full and commands were permanently lost.	Event
7	Time-out error. system did not maintain minimum ramp rate	Event

state of the item they report at the time the Status Byte was read. Event bits indicate that an event took place since the last time the Status Byte was read. Bits 0, 1, 2, 3, 5, 6, and 7 are event bits. Bit 4 is a state bit.

State bits are set and reset by the condition they report. Thus, if any of the bits in bytes 02-31 of the Error/Status String are set, bit 1 will be set. If all of the bits in bytes 02-31 of the Error/Status String are reset, bit 4 of the Error Byte will be reset. Reading the Error Byte does not affect the state of bit 4.

Event bits are set by the condition they report. However, they are only reset when the Error Byte is read or the controller receives a Device Clear, or is powered off. Thus, an event bit tells you if an event took place since the last time you read the Error Byte. There are a number of methods to “read” the Error Byte. The RE and REA commands read the Error Byte directly. The QE and QEA commands read the Error Byte as part of the Error/Status String.

Except for bit 4, the Error Byte is “event” information. Each time the state of an event item that is reported in the Error Byte becomes true, the associated event bit in the Error Byte is set. If the condition of the reported event item subsequently becomes false, the associated event bit is not reset. The entire Error Byte, except bit 4, is cleared (reset) after it is read with the RE, REA, QE, or QEA commands. As the Error Byte is byte 00 of the Error/Status String, byte 00 of the Error/Status String is cleared when the Error Byte is read.

Bit 0 Aux Input Toggled

Each time the state of the TTL Aux Input changes, from open to closed, or from closed to open, bit 0 will be set. Note: The current state of the Aux Input can be read from the Error/Status String, byte 35, bit 7.

Bit 1
System Fault Alert

If the system is equipped with the optional Fault Sensing System, and that system detects a monitored component failure, bit 1 will be set. Because this bit represents the “ORing” of the various component fault signals of the Component Fault Monitoring System, bit 1 will not be set again until the system has run without a fault condition. That is, if the Error Byte is read after a fault indication, bit 1 will be reset. If system operation is continued without resolving the condition that is reported by the fault indication, then neither the continuing fault, nor any additional faults will be reported. The Component Fault Monitoring System expects faults to be cured and the system to operate from a fault-free (logical 0) base condition.

Bit 2
Illegal Argument String

This bit is set when a command is received from the host controller that is a valid command, but which is accompanied by an argument (parameter) that is not in the specified format or that is not within the legal range. The command and the argument data will be discarded.

Bit 3
Fail-safe Tripped

Bit 3 is set when the fail-safe system (either bi-metal or precision) shuts down the system due to an over or under temperature condition. See §3.10 for a detailed discussion of failsafe operation.

Bit 4
Error/Status String Data

Bit 4 is set when any of the “event” bits (bytes 02-31) in the Error/Status String are set. It is incumbent upon the user to read the Error/Status String after reading the Error Byte to ascertain the nature of the reported event and to clear the event section of the Error/Status String. It is important to recognize that bit 4 of the Error Byte is the “ORed” sum of bytes 02-31 of the Error/Status String, and is thus cleared (reset) only when all bits in bytes 02-31 of the Error/Status String are reset.

Bit 5
Undefined Command

Bit 5 is set when the controller receives a command from the host computer that it does not recognize. The controller features a “fault tolerant” parser that will try to fix syntax errors, line termination, etc. However, if the parser cannot find a known command at the beginning of a string of received data, bit 5 will be set and the data received will be discarded.

Bit 6
Command Buffer Overrun

Bit 6 is set when the controller receives commands it cannot store because the command buffer is full. The data received will be discarded. Note: Bits 1 & 2 of the Status Byte provide for a method of flow control to prevent this problem. See §7.3.4 on communications flow control.

Bit 7
Time-out Error

Bit 7 is set when the device being controlled fails to achieve temperature transitions at the minimum rate specified in Setup Parameter F33.

Status Byte

Introduction

The Status Byte is a one byte block of data in which each of the eight bits reports the status of an item. Each bit can be either 0 or 1. A bit with a value of 0 is “reset” and is equivalent to logical false. A bit with a value of 1 is “set” and is equivalent to logical true. The bits in the Status Byte report the status of the following items:

Bit	Definition - Status Byte	Type	Sets SRQ
0	SRQ enabled (1), disabled (0)	State	No
1	Buffer over at least 75% full - stop sending	State	Yes
2	Buffer not more than 50% full - resume sending	State	Yes
3	Interval complete	Event	Yes
4	Set point reached	Event	Yes
5	Run mode active (controlling temperature)	State	No
6	SRQ set	Event	--
7	Error	State	Yes

Some of the Status Byte bits are “state” bits and others are “event” bits. State bits reflect the state of the item they report at the time the Status Byte was read. Event bits indicate that an event took place since the last time the Status Byte was read. Bits 0, 1, 2, and 7 are state bits. Bits 3, 4, & 6 are event bits.

State bits are set and reset by the condition they report. Thus, if the buffer is at least 75% full, bit 1 will be set. If the buffer is not at least 75% full, the bit will be reset. Reading the Status Byte does not affect the state bits.

Event bits are set by the condition they report. However, they are only reset when the Status Byte is read or the controller receives a Device Clear, or is powered off. Thus, an event bit tells you if an event took place since the last time you read the Status Byte. There are a number of methods to “read” the Status Byte. The RS and RSA commands and the GPIB serial poll (spool) command read the Status Byte directly. The QE and QEA commands read the Status Byte as part of the Error/Status String. The Status Byte is also deemed read when it is sent in RS-232 mode as a result of having SRQ enabled (see below).

SRQ is a special function of the Status Byte. Setting of bit 6, the SRQ bit, indicates that the Status Byte has new information and needs to be read and processed. If the SRQ function has been enabled by the ES command, then the SRQ will be “asserted”. If the communication bus is IEEE-488, the SRQ line will be set high to signal to the bus controller that the Status Byte needs to be read. If the communication bus is RS-232, the Status Byte will automatically be sent. See §7.8.1 for a complete discussion on reading the Status Byte.

The Status Byte is contained in byte 01 of the Error/Status String.

Following is an explanation of the function and behavior of each of the bits in the Status

Byte:

Bit 0 SRQ Enabled

Bit 0 is set when the SRQ function is enabled by the ES command. Bit 0 is reset when the SRQ function is disabled at power on or by the DS command. See §7.8.3 for a discussion of the SRQ Function. Bit 0 is not reset when the Status Byte is read. Setting bit 0 does not assert the SRQ, it simply sets the controller in the mode to assert the SRQ when bit 6 is set.

Bit 1
Bit 2
Serial Mode Buffer at least x% full

Bit 1: Serial Mode Buffer at least 75% Full.
Bit 2: Serial Mode Buffer at least 50% Full.

These bits are for serial communications flow control. They are only set when data is being received via the serial port. These bits are not used when data is being sent via GPIB as the GPIB has its own flow control.

If you have used the SP command to select Remote Mode programming of the controller, the controller can receive and store commands to be executed. If the buffer that stores the commands becomes 75% full, bit 1 will be set indicating that transmissions to the controller should pause. When enough commands have been taken from the buffer that the buffer is less than 50% full, bit 2 will be set indicating that there is now enough room in the controller buffer for command transmission to resume. See Communication Flow Control, page 3-14 for information on communications flow control. Bits 1 & 2 are not reset when the Status Byte is read. Bit 1 is reset when the buffer is less than 75% full. Bit 2 is reset when the buffer is more than 50% full. Setting either bit 1 or bit 2 sets the SRQ, bit 6, as well.

Note: these bits are set for GPIB communications as well. Their significance in GPIB mode is simply to note that the bits may change. As the GPIB has its own flow control, the setting/resetting of these bits in GPIB mode is insignificant.

Bit 3
Interval Complete

Bit 3 is set when a timed delay (DL command) or ramping interval (RA, RAF commands) is completed. Bit 3 is reset when the Status Byte is read. Setting bit 3 sets the SRQ, bit 6, as well.

Bit 4
Set point Reached

Bit 4 is set when a go to set point (GT, GTF commands) is satisfied. A set point is satisfied when the system temperature has remained stable about the set point within the limits prescribed by settling band parameters (F31 & F32). For more information refer to SB, SBF Set Settling Band Parameters, page 3-27. Bit 4 is reset when the Status Byte is read. Setting bit 4 sets the SRQ, bit 6, as well.

Bit 5
Run Mode

Bit 5 is set when the controller is actively controlling the temperature. When the controller is idle, bit 5 is reset.

Bit 6
SRQ Set

Bit 6 is set whenever bits 1, 2, 3, 4, or 7 are set. Bit 6 is reset when the Status Byte is read. If the SRQ is enabled (see bit 0), the host controller will be notified as follows that the SRQ (bit 6) is set:

Bit 7
Error

Bit 7 is set when any bit is set in the Error Byte. Every function that sets a bit in the Error Byte, also sets bit 7 and bit 6 of the Status Byte. Bit 7 is reset only when the Error Byte is read and thus cleared.

Error/Status String

Introduction

The Error/Status String is a string of 64 bytes (512 bits) that can be obtained from the controller. Not all of the bytes in the Error/Status String are currently defined. Although the information contained in the Error/Status String is at the bit level, related information is grouped by bytes. Specific information can then be obtained by selecting the needed information group byte by offset. The bit level information can then be used or extracted.

String Organization

The bytes of the Error/Status String are arranged in three groups:

Bytes	Purpose
00 - 01	Error Byte & Status Byte
02 - 31	Event Information
32 - 63	Status Information

Some of the Error/Status String bits are “state” bits and others are “event” bits. Event bits indicate that an event took place since the last time the Error/Status String was read. State bits reflect the state of the item they report at the time the Error/Status String was read. Bits in bytes 02-31 are event bits. Bits in bytes 32-63 are state bits.

Event bits are set by the condition they report. However, they are only reset when the Error/Status String is read or the controller receives a Device Clear, or is powered off. Thus, an event bit tells you if an event took place since the last time you read the Error/Status String.

State bits are set and reset by the condition they report. Thus, if the refrigeration compressor is turned on, byte 35, bit 3 will be set. If the compressor is turned off, the bit will be reset. Reading the Error/Status String does not affect the state bits.

If a bit is set (1), the condition is true, if a bit is reset (0), the condition is false.

Note: Byte 00 of the Error/Status String is the Error Byte that is available using RE and REA commands. Byte 00 is cleared (all bits reset) each time a QE or QEA command is issued. This byte is also cleared anytime the RE or REA commands are issued to read the Error Byte. This byte is maintained in the Error/Status String so that it is not necessary to use two commands to obtain the Error Byte and the extended error information contained in the Error/Status String.

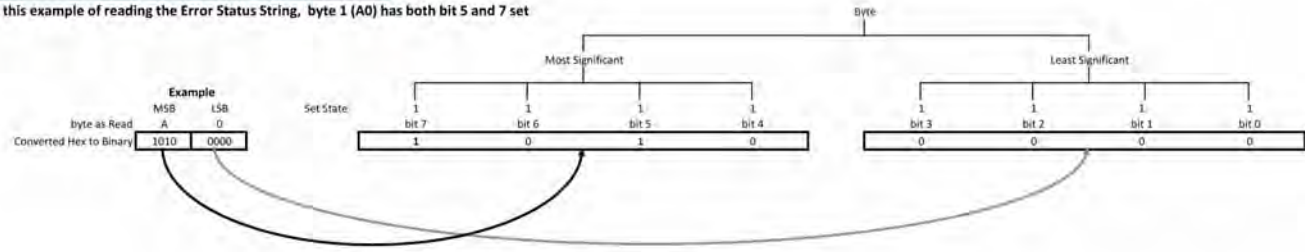
Byte 01 of the Error/Status String is the Status Byte that is available using the RS, RSA, and serial poll commands. Bits 3, 4, & 6 are cleared (reset) when the Status Byte or the Error/Status String (containing the Status Byte) is read.

Many of the bits in the Error/Status String have no current definition and will always be reset (0). Successive releases of Sigma Systems controller firmware may implement additional bit definitions. The length of the Error/Status String will remain at 64 bytes.

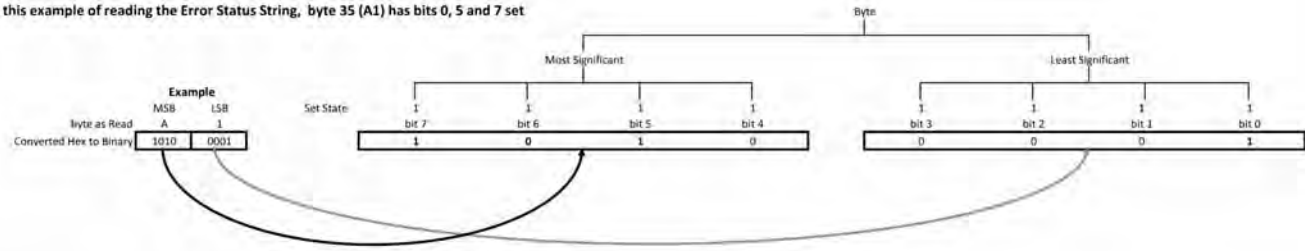
Error String Examples

Use the following examples to understand the error string.

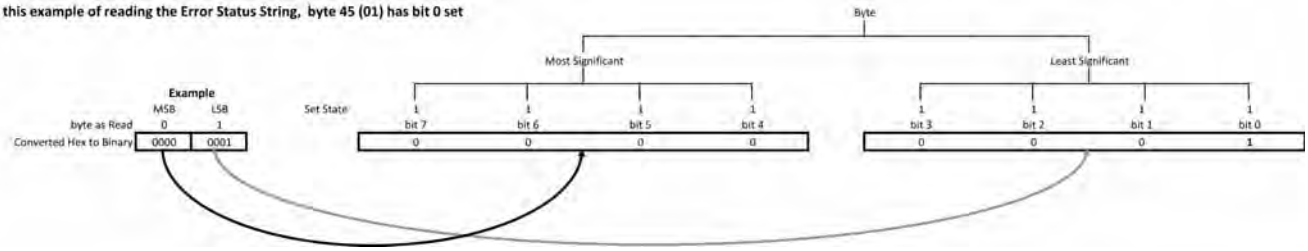
In this example of reading the Error Status String, byte 1 (A0) has both bit 5 and 7 set



In this example of reading the Error Status String, byte 35 (A1) has bits 0, 5 and 7 set



In this example of reading the Error Status String, byte 45 (01) has bit 0 set



Error/Status String Bit Definitions

ERROR BYTE & STATUS BYTE

Byte 00 - Error Byte

00:0 Aux input toggled (typically optional door switch open/closed)

00:1 Fault system alert

00:2 Illegal parameter (A valid command was received with an illegal parameter)

00:3 Fail-safe tripped

00:4 Event set in bytes 02-31 of Error/Status String

00:5 Unrecognized command (the controller received data that it did not recognize)

00:6 Command buffer overrun (data was received and lost because the buffer was full)

00:7 Set point Timeout error (specified minimum ramp rate was not maintained)

Byte 01 - Status Byte

01:0 SRQ enabled

01:1 Buffer at least 75% full - stop sending commands

01:2 Buffer not more than 50% full - resume sending commands

01:3 Interval complete

01:4 Set point reached

01:5 Run mode active (controlling temperature)

01:6 SRQ set

01:7 Error (bit set in Error Byte)

Event Bytes

Byte 02 - Set point, Probe & Command Errors

02:0 Set point submitted exceeds SOR upper operating limit

02:1 Set point submitted exceeds SOR lower operating limit

02:2 Probe requested is not defined in F1

02:3 reserved

02:4 reserved

02:5 Not used

02:6 RA command received while PN=0

02:7 RR command received while PN=0

Byte 03 - Probe Temperature Value Errors

03:0 Probe 1 more than 20°C and less than 50°C above high temp operation limit

03:1 Probe 1 more than 20°C and less than 50°C below low temp operation limit

03:2 Probe 2 more than 20°C and less than 50°C above high temp operation limit

03:3 Probe 2 more than 20°C and less than 50°C below low temp operation limit

03:4 Probe 1 more than 50°C above high temp operation limit

03:5 Probe 1 more than 50°C below low temp operation limit

03:6 Probe 2 more than 50°C above high temp operation limit

03:7 Probe 2 more than 50°C below low temp operation limit

Byte 04 - Correction Value Errors

04:0 Correction temperatures submitted less than 50°C apart

04:1 Correction temperatures submitted resulted in a correction greater than 10% of the controller range

at the limits of that range

04:2 Correction temperatures submitted were outside the System Operating Range

Byte 05 - Hardware & Setup Errors

05:0 EEPROM write failed

05:1 FLASH memory write failed

05:2 F25 >26 or F27>F28 or lower SOR limit > upper SOR limit

05:3 Not used

05:4 Probe 1 open

05:5 Probe 1 shorted

05:6 Probe 2 open

05:7 Probe 2 shorted

Byte 06 - Control Events

06:0 reserved

31.x Internal system errors

(if any bit in byte 31 gets set, notify Sigma Systems technical support)

State Bytes

Byte 35 - Port status (set=on, reset=off)

35:0 Blower port on

35:1 Main heater port (control heat) on

35:2 Boost heater port on

35:3 Refer (compressor) port on

35:4 Aux output (power) port on

35:5 Coolant port on

35:6 Fault alert

35:7 Aux input port on (connected to ground)

Byte 36 - Probe Temperature Out Of Range State

36:0 Probe 1 more than 20 C and less than 50 C above high temp operation limit

36:1 Probe 1 more than 20 C and less than 50 C below low temp operation limit

36:2 Probe 2 more than 20 C and less than 50 C above high temp operation limit

36:3 Probe 2 more than 20 C and less than 50 C below low temp operation limit

36:4 Probe 1 more than 50 C above high temp operation limit

36:5 Probe 1 more than 50 C below low temp operation limit

36:6 Probe 2 more than 50 C above high temp operation limit

36:7 Probe 2 more than 50 C below low temp operation limit

Byte 37 - Probe Temperature Out of Range & Failsafe State

37:0 GPIB Command overrun

37:1 GPIB General error.

37:2 GPIB Timeout error

37:3 Transient Fail-safe occurred

37:4 Probe 1 open

37:5 Probe 1 shorted

37:6 Probe 2 open

37:7 Probe 2 shorted

Byte 45 - Port Control Flags

45:0 Blower enabled (when controller idle - not controlling temperature)

Byte 46 - Control States

46:0 reserved

Section I: Reset Commands

Introduction

The controller uses one common reset routine to reset the system. The multiple reset levels required by SCPI and IEEE-488.2 are not supported. *RST is not supported. The controller command DC, the GPIB commands Device Clear and Selected Device Clear all cause the common reset routine to be executed. The reset routine performs the following functions:

Flush the communications input & ring buffers
Flush the command buffer
Clear the GT/RA command time-out flag
Turn the refrigeration compressor port to its default state per setup parameter F6
Turn the Aux/Power Port to its default state per setup parameter F5
Clear the Aux input flag
Clear the Component Fault input flag
Clear the event bits in the Error Byte, Status Byte, and Error/Status string
Setup the Serial Poll register
Reset the SRQ enable bit
Turn controller off (stop all heating and cooling, turn off blowers)
Return the set point to 30°C
Replaces any temporary setup parameters with the values from the permanent table

Note: Beginning with firmware version 8.6.3, Interface Clear does NOT trigger the reset routine

Maintenance & Cleaning

4

Overview

The ThermoSpot is designed to operate maintenance-free for long durations. There are no filters, strainers, or expendables requiring maintenance.

Cleaning the Condenser Fins:

The most critical maintenance procedure is cleaning the condenser fins of the refrigeration system. The front and rear grill of the system should be vacuumed monthly (or more frequently) to remove any dust and debri buildup.

Air mus flow freely through the chassis and dust can restrict that air flow causing reduction in performance and eventually, component failure.

If the dust buildup can't be removed by vacuuming, power down the system, disconnect from facility power and air, remove the cover and use clean, dry compressed air to blow the dust and debris out of the fins and chassis.

Cleainig the Touch-screen

Use and LCD safe cleaner and cloth to wipe the screen and chassis clean.

Inspect Power Cable and Purge Air Lines

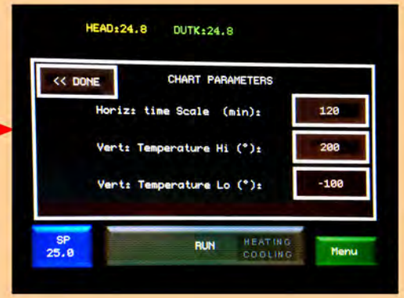
Before using the system, inspect the power cable to verify that there are no cuts or nicks on the cable.
Inspect the Purge Aur Lines, verify that there are no kins or restrictions in the lines.

Replace any damaged power or air lines.

DCP-Series Controller Screen Tree

Start

Graph

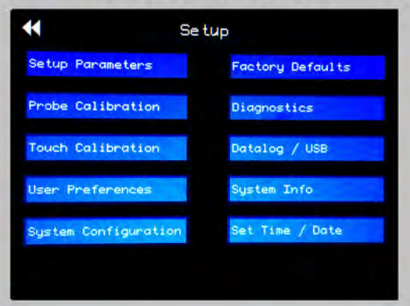


Local Mode

Program Mode

Program Mode Details On Page 2 of 3

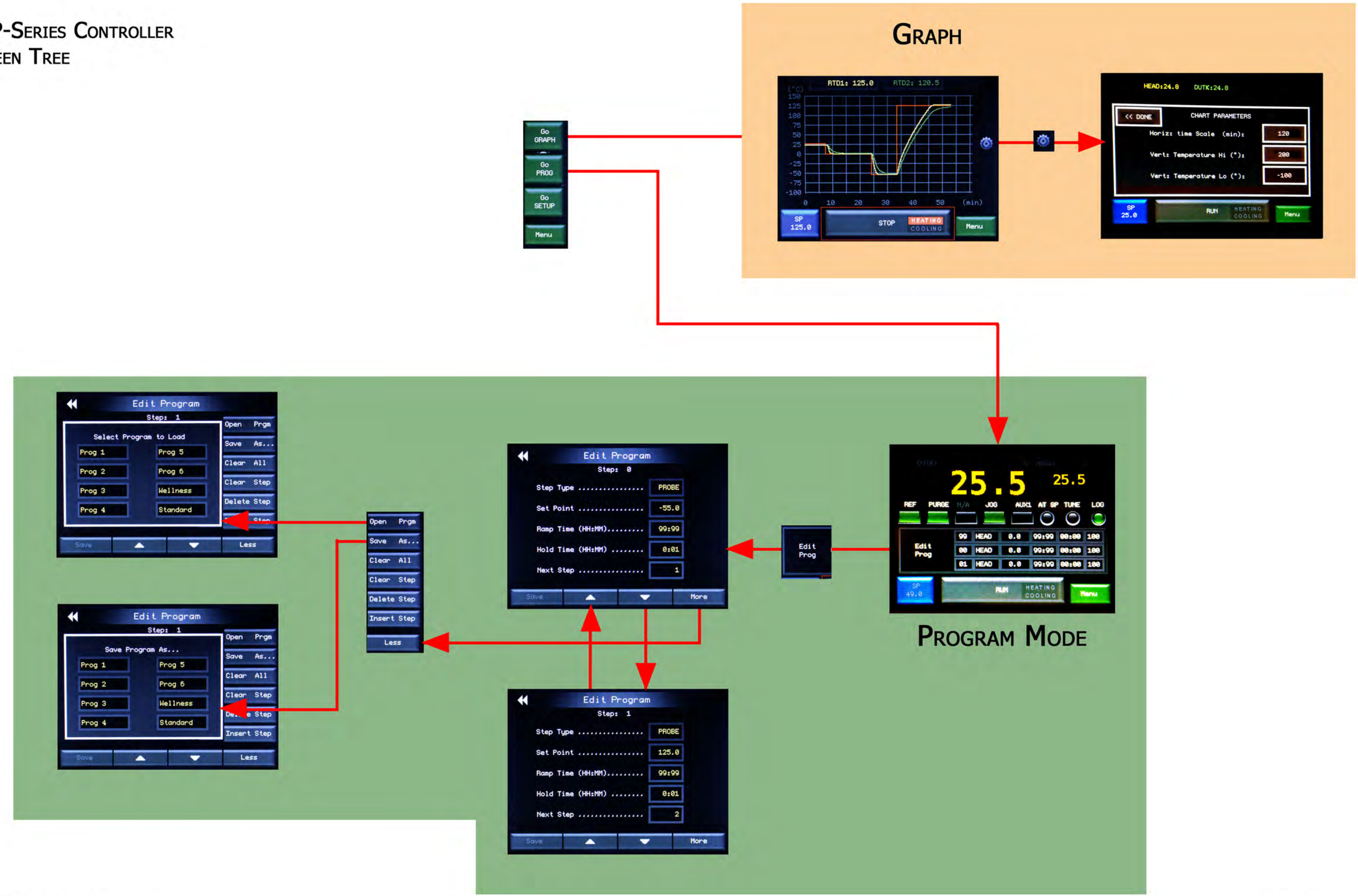
Setup Mode



Setup Mode Details On Page 3 of 3

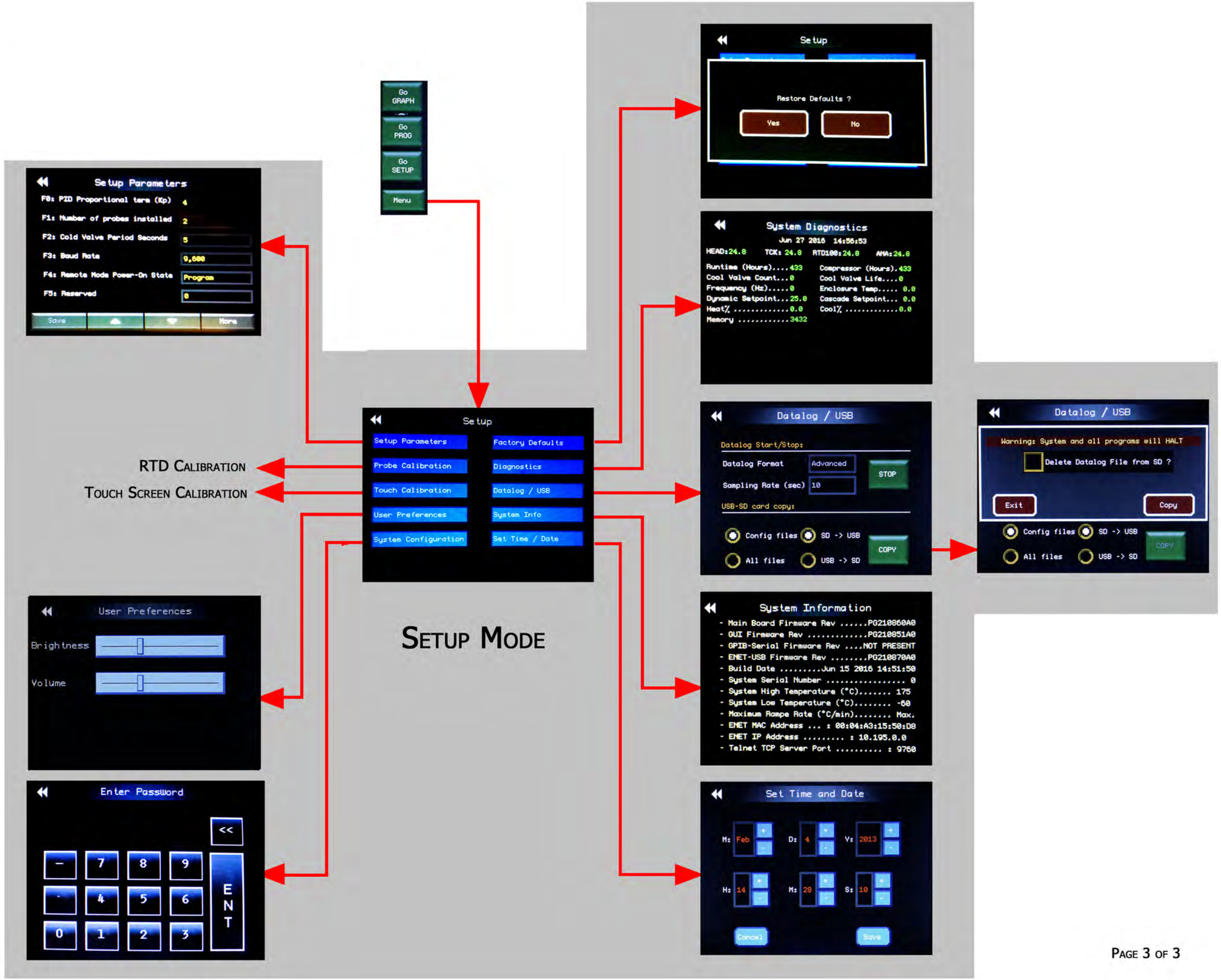
Universal Back Button

DCP-Series Controller Screen Tree



Universal Back Button

DCP-Series Controller Screen Tree



Universal Back Button